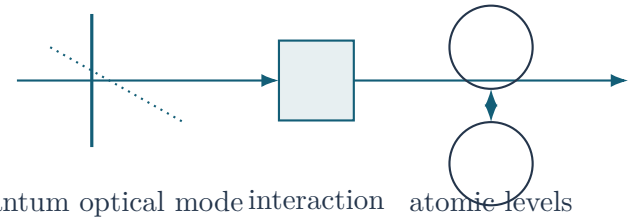
PHYS4055

Atomic Physics and Quantum Optics

A Mini-Textbook

From physical ideas to equations and worked examples



Four connected parts:
I. Quantised light, photon statistics and squeezing
II. Quantum information, open systems and metrology
III. Atomic wavefunctions, electron coupling and structure
IV. Spectroscopy, cooling, symmetry tests and atomic clocks

Course-based revision edition · November 2026

Companion to the 15-page overview, tutorial/assignment worked solutions and 2023–2025 exam answers.

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How to use this mini-textbook

This is a *teaching document* for the full PHYS4055 course, not an abbreviated formula sheet. Its purpose is to answer three questions: what is happening physically, why the formula has its particular form, and how that formula helps solve an exam problem. The 15-page overview is designed for fast lookup; this book is intended to be *read and worked through*.

A suggested route through the material

The course has four connected parts. Quantum optics first describes light as quantised field modes. Applications of optics use those states, operations and measurements to communicate, process information and make sensitive measurements. Atomic physics introduces electron wavefunctions, angular momentum, energy-level splitting and transition matrix elements. Applications of atomic physics use these ideas in cooling and trapping, symmetry tests and clocks.

If quantum notation feels unfamiliar, start with Chapters 1–4 before trying the applications. If your main difficulty is angular-momentum coupling, read Chapters 14–16 slowly, including their worked examples. Later chapters repeatedly refer back to these ideas.

Mathematical notation, explained

A ket $|\psi\rangle$ is a quantum state vector. A bra $\langle\phi|$ is its conjugate transpose. The inner product $\langle\phi|\psi\rangle$ is a complex amplitude and $|\langle\phi|\psi\rangle|^2$ is a probability when the states are normalised. A matrix element $\langle f|O|i\rangle$ asks how strongly the operator O connects initial $|i\rangle$ to final $|f\rangle$. A commutator is $[A, B] = AB - BA$; it measures the difference caused by operator order.

A density operator ρ describes both pure states and mixtures. An expectation is $\langle O\rangle = \mathrm{Tr}(\rho O)$. A partial trace sums over outcomes of a subsystem that is not observed. A unitary $U^\dagger U = I$ preserves state norms. A Hermitian observable $O = O^\dagger$ has real measurement eigenvalues.

Worked example: How to read a transition matrix element

The expression $\langle 3s|\,\mathbf{d}\,|3p\rangle$ does not directly give a transition probability. It gives a *dipole amplitude*, including radial overlap and angular selection rules. Its squared magnitude enters an Einstein A rate, multiplied by a factor proportional to the cube of transition frequency. Only after summing all decay channels can we infer a lifetime.

Conventions we maintain

For the quantum-optics chapters,

$$x = a + a^\dagger, \qquad p = i(a^\dagger - a), \qquad [x, p] = 2i,$$

so vacuum quadrature variance is 1. Optical beam-splitter phases are defined locally in each example; physical probabilities are invariant under a consistent convention change.

For atomic physics, lower-case l, s often refer to one electron, capital L, S to the combined electrons, J to electronic total angular momentum, I to nuclear spin and F to their sum. The two symbols m_J, m_F name projections along the chosen quantisation axis.

The four checks to do at the end

1. **Probability:** do all output probabilities sum to 1? Is $\mathrm{Tr}\,\rho = 1$?
2. **Symmetry:** is the transition allowed by triangle, magnetic projection and parity conditions?
3. **Physical limiting case:** what happens if the loss is zero, the detuning is zero, or the field is removed?

4. **Units:** do the answers have the expected dimensions and order of magnitude?

Each chapter ends with a short concept check. For longer algebraic practice, use the companion Worked Examples, Assignment Solutions and Past Exam Solutions booklets in the repository.

1 Why light is a quantum oscillator

Course sources: Week 1 lectures; Week 1 tutorial

The physical picture

An optical photon is one excitation of an *electromagnetic mode*. The quantum theory does not replace Maxwell's waves with little balls: it quantises the energy and fluctuations of each independent wave mode.

1.1 From Maxwell's equations to a mode

A classical electromagnetic field can be expanded in spatial and polarisation modes. A cavity enforces standing-wave boundary conditions; free space supports travelling-wave modes. After suitable normalisation, the energy in one classical mode has the harmonic-oscillator form

$$H = \tfrac{1}{2}(p^2 + \omega^2 q^2).$$

Quantisation makes q, p operators satisfying $[\hat{q}, \hat{p}] = i\hbar$. Define

$$\hat{a} = \sqrt{\frac{\omega}{2\hbar}}\,\hat{q} + \frac{i\hat{p}}{\sqrt{2\hbar\omega}}, \quad \hat{a}^\dagger = \sqrt{\frac{\omega}{2\hbar}}\,\hat{q} - \frac{i\hat{p}}{\sqrt{2\hbar\omega}}.$$

Substitution gives the central result

$$\boxed{[\hat{a}, \hat{a}^\dagger] = 1, \qquad H = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \tfrac{1}{2}\right).}$$

The zero-point term $\hbar\omega/2$ describes the energy left when the oscillator cannot have both conjugate coordinates exactly zero.

1.2 What raising and lowering actually do

The number operator $\hat{n} = \hat{a}^\dagger\hat{a}$ has eigenstates $|n\rangle$ with integer eigenvalues:

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle\,.$$

The lowest state is vacuum, $\hat{a}\,|0\rangle = 0$. An arbitrary n-photon state can be built by

$$\boxed{|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle\,.}$$

The factorial is required for normalisation; it becomes especially important with two or more photons.

An observable optical electric field contains both annihilation and creation parts:

$$\hat{E}(z,t) = E_0[\hat{a}e^{i(kz-\omega t)} + \hat{a}^\dagger e^{-i(kz-\omega t)}].$$

Although $|n\rangle$ contains energy $n\hbar\omega$ above vacuum, $\langle n|\hat{a}|n\rangle = 0$. It therefore has zero *mean electric field*. A nonzero mean photon number does not imply a classically oscillating mean wave.

1.3 Several modes and distinguishability

For two modes a, b, $|n, m\rangle$ means n photons in a, m in b. Examples are

$$|1,1\rangle = a^\dagger b^\dagger\,|0,0\rangle\,, \qquad |2,0\rangle = \frac{(a^\dagger)^2}{\sqrt{2}}\,|0,0\rangle\,.$$

A mode label may describe path, polarisation, frequency, temporal envelope or a combination. Two photons become distinguishable when a measurement could, in principle, identify which came from which input (even if nobody actually reads that information). Their *mode overlap* therefore controls interference.

Worked example: Why $\sqrt{2!}$ matters

Compute $\|(a^\dagger)^2 |0\rangle\|^2 = \langle 0| a^2 (a^\dagger)^2 |0\rangle = 2$. So $(a^\dagger)^2 |0\rangle$ is *not* a unit vector; dividing by $\sqrt{2}$ produces $|2\rangle$. This factor later changes beam-splitter output amplitudes and probabilities.

1.4 What a phase shifter does

Heisenberg evolution under the free oscillator Hamiltonian is

$$\frac{d\hat{a}}{dt} = \frac{i}{\hbar}[H, \hat{a}] = -i\omega\hat{a}, \qquad \hat{a}(t) = e^{-i\omega t}\hat{a}(0).$$

The photon number does not change: the mode simply accumulates phase. In an interferometer, *relative* phases become measurable after different paths are recombined.

Concept check

Three different objects: a mode specifies which oscillator; a photon is one excitation of that oscillator; a quantum state describes the probabilities and coherences among its possible excitation numbers.

2 Coherent, number and thermal light

Course sources: Weeks 1–2; photon statistics tutorial; Assignment 1

2.1 What does classical-looking light mean?

A coherent state satisfies $\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle$ for a complex amplitude $\alpha = |\alpha|e^{i\theta}$. Unlike a number state, it is a *superposition* of different photon numbers:

$$\boxed{|\alpha\rangle = e^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle\,.}$$

The probability of n photons is obtained by squaring the corresponding amplitude:

$$P(n) = e^{-|\alpha|^2}\frac{|\alpha|^{2n}}{n!}.$$

This is Poissonian: $\langle n\rangle = |\alpha|^2$, $\mathrm{Var}(n) = |\alpha|^2$. A coherent state has a mean oscillating electric field because $\langle a\rangle = \alpha$, with $\langle E\rangle \propto 2|\alpha|\cos(kz - \omega t + \theta)$.

2.2 Displacement: making a coherent state

The unitary displacement operator

$$D(\alpha) = e^{\alpha a^\dagger - \alpha^* a}$$

acts on vacuum to give $|\alpha\rangle = D(\alpha)\,|0\rangle$. A useful calculation employs the BCH identity:

$$D^\dagger a D = a + [a, \alpha a^\dagger - \alpha^* a] + \tfrac{1}{2}[[a, \alpha a^\dagger - \alpha^* a], \ldots] = \boxed{a + \alpha}.$$

The first commutator equals α, a scalar; all higher nested commutators vanish. Therefore $aD\,|0\rangle = D(a+\alpha)\,|0\rangle = \alpha D\,|0\rangle$, which proves the eigenstate property.

2.3 A density operator describes mixtures

A pure state $|\psi\rangle$ has $\rho = |\psi\rangle\langle\psi|$. When preparation involves several alternative states,

$$\boxed{\rho = \sum_j p_j\,|\psi_j\rangle\langle\psi_j|\,, \quad \langle O\rangle = \mathrm{Tr}(\rho O).}$$

A mixture differs from a coherent superposition. For example, $(|0\rangle + |1\rangle)/\sqrt{2}$ has off-diagonal density-matrix terms, whereas a 50–50 mixture of $|0\rangle, |1\rangle$ does not. Both have the same probabilities in a photon-number measurement, but different phase-sensitive expectations. Purity is measured by $\mathrm{Tr}\,\rho^2$: it equals 1 for a pure state and is smaller for a mixture.

2.4 Thermal light and bunching

A single thermal optical mode has

$$\rho_{\mathrm{th}} = \sum_{n=0}^{\infty}\frac{\bar{n}^n}{(1+\bar{n})^{n+1}}\,|n\rangle\langle n|\,, \quad \bar{n} = \frac{1}{e^{\hbar\omega/(k_B T)} - 1}.$$

The variance is $\mathrm{Var}(n) = \bar{n} + \bar{n}^2$. Compare with coherent light, where the variance equals the mean, and a fixed-n Fock state, where the variance vanishes. The Mandel parameter

$$Q_M = \frac{\mathrm{Var}(n) - \bar{n}}{\bar{n}}$$

describes super-Poissonian ($Q_M > 0$), Poissonian ($Q_M = 0$) and sub-Poissonian ($Q_M < 0$) fluctuations.

Worked example: Three states with mean photon number 4

A coherent state with $|\alpha|^2 = 4$ has variance 4. A thermal mode with $\bar{n} = 4$ has variance $4+16 = 20$. The Fock state $|4\rangle$ has variance 0. Equal mean energy does *not* mean equal noise.

2.5 Second-order coherence counts pairs

The zero-delay Glauber correlation is

$$\boxed{g^{(2)}(0) = \frac{\langle a^{\dagger 2} a^2 \rangle}{\bar{n}^2} = \frac{\langle n(n-1) \rangle}{\bar{n}^2}}.$$

Why $n(n-1)$? It measures the number of ordered pairs that can be formed from n photons, which is relevant for two detections. A one-photon state has $g^{(2)} = 0$: it cannot cause two ideal simultaneous counts. A coherent state has $g^{(2)} = 1$; a thermal mode has $g^{(2)} = 2$.

2.6 Entangled pairs can look thermal locally

A two-mode squeezed state has the form

$$|\Psi\rangle = \sqrt{1-\lambda^2} \sum_n \lambda^n \, |n\rangle_s \, |n\rangle_i \, .$$

It has perfectly correlated photon numbers. Tracing over the unobserved idler gives

$$\rho_s = (1-\lambda^2) \sum_n \lambda^{2n} \, |n\rangle \langle n| \, , \quad \bar{n} = \frac{\lambda^2}{1-\lambda^2}.$$

This is *thermal locally* despite the joint state being pure. The signal–idler cross-correlation is

$$g^{(2)}_{si}(0) = \frac{\langle n_s n_i \rangle}{\bar{n}^2} = 2 + \frac{1}{\bar{n}}.$$

At low gain, detection of one idler photon heralds one signal photon.

Concept check

A photon-number histogram alone cannot reveal all coherences or entanglement. Ask whether the experiment measures number, field phase, or correlations between modes.

3 Beam splitters, HOM interference and detectors

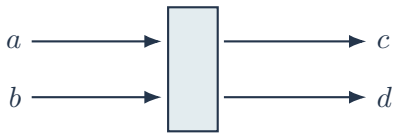
Course sources: Weeks 2–3; optics tutorials; 2024–25 exams

3.1 How an interferometer mixes amplitudes

A lossless two-port beam splitter transforms field *operators*, not photon probabilities:

$$\boxed{\begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} t & r \\ -r & t \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}, \quad t^2 + r^2 = 1.}$$

The numbers t, r are amplitudes; the intensity fractions are $T = t^2, R = r^2$. The unitary matrix preserves both photon number and bosonic commutation relations. Reflection phases may differ in another convention, with identical observable predictions.



unitary mixing

3.2 Multiphoton calculations use creation operators

When photons are input in known number states, write the input with raising operators. For the balanced convention

$$a^\dagger = \frac{c^\dagger + d^\dagger}{\sqrt{2}}, \qquad b^\dagger = \frac{c^\dagger - d^\dagger}{\sqrt{2}},$$

the transformation of $|1,1\rangle$ is

$$\begin{aligned} a^\dagger b^\dagger |0\rangle &= \tfrac{1}{2}(c^\dagger + d^\dagger)(c^\dagger - d^\dagger)|0\rangle \\ &= \tfrac{1}{2}[(c^\dagger)^2 - (d^\dagger)^2]|0\rangle \\ &= \boxed{\frac{|2,0\rangle - |0,2\rangle}{\sqrt{2}}}. \end{aligned}$$

The absence of a $|1,1\rangle$ term means zero probability of detecting one photon in each output. This is Hong–Ou–Mandel (HOM) interference.

3.3 Where the HOM cancellation comes from

Two indistinguishable possibilities lead to a coincidence: both photons transmit, or both reflect. Their probability *amplitudes* have opposite signs and cancel at a balanced splitter. If the photons have distinguishable wavepackets, those alternatives no longer interfere completely. For ideal overlap parameter $\eta \in [0,1]$, defined as the squared temporal/spectral-mode overlap,

$$P(\text{coincidence}) = \frac{1-\eta}{2}.$$

At $\eta = 0$, the classical independent-path coincidence probability is $1/2$. At $\eta = 1$, HOM bunching gives zero.

Worked example: Two photons enter the same port

For $|2,0\rangle = (a^\dagger)^2|0\rangle/\sqrt{2}$,

$$|2,0\rangle \longrightarrow \tfrac{1}{2}|2,0\rangle + \tfrac{1}{\sqrt{2}}|1,1\rangle + \tfrac{1}{2}|0,2\rangle.$$

The output probabilities are $1/4, 1/2, 1/4$. For $|1,1\rangle$ input, by contrast, the coincidence probability is *zero*. Different input states lead to different interference and photon-counting patterns.

3.4 Arbitrary n photons and coherent input

A fixed n-photon input gives the binomial result

$$|n,0\rangle \mapsto \sum_{k=0}^{n} \sqrt{\binom{n}{k}}\, t^k r^{n-k}\, |k, n-k\rangle\,.$$

A coherent input $|\alpha\rangle_a\,|0\rangle_b$ instead produces a *product* of coherent outputs $|t\alpha\rangle_c\,|r\alpha\rangle_d$ (up to phases). This follows by substituting the creation-operator transformation into the exponential Fock expansion.

3.5 What a detector measures

An ideal number-resolving detector distinguishes $n = 0, 1, 2, \ldots$. A threshold detector only reports vacuum or at least one photon. For n incident photons and single-photon detection efficiency η_d, the probability of *no click* is

$$P(0 \text{ clicks} \mid n) = (1-\eta_d)^n,$$

neglecting dark counts. Real detection requires careful treatment of loss, dark counts and finite time resolution.

3.6 Homodyne detection accesses field quadratures

Interfere a weak signal a with a strong coherent local oscillator $\beta = |\beta|e^{i\theta}$. Subtract the output photodetector currents. The dominant difference term is proportional to

$$\beta^* a + \beta a^\dagger = |\beta|[a e^{-i\theta} + a^\dagger e^{i\theta}].$$

A local-oscillator phase change selects a different quadrature rather than a different photon-number observable.

Concept check

For optical circuits: define input modes, choose a phase convention, transform operators, restore the $\sqrt{n!}$ factors, add amplitudes, and only then square them.

4 Quadratures and phase-space pictures

Course sources: W1L3 and W3L1; homodyne lectures

4.1 What are the two quadratures of a wave?

A monochromatic classical electric field can be written

$$E(t) = A\cos\omega t + B\sin\omega t.$$

The coefficients A, B specify its amplitude and phase. They are not frequency and amplitude: the *frequency* ω is already fixed by the mode. Quantum mechanically, define

$$\boxed{x = a + a^\dagger, \qquad p = i(a^\dagger - a), \qquad a = (x + ip)/2.}$$

The commutator is

$$[x, p] = [a + a^\dagger, i(a^\dagger - a)] = 2i.$$

Heisenberg's inequality gives

$$\boxed{\Delta x\, \Delta p \geq 1.}$$

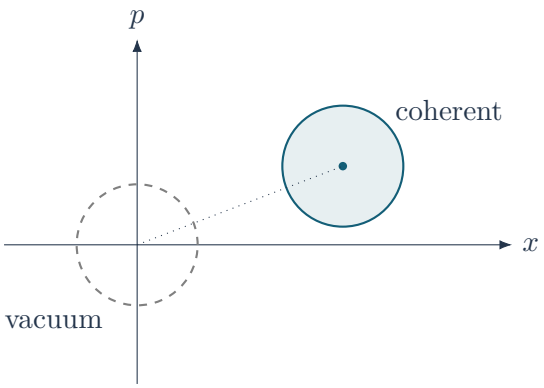
In this course convention the vacuum has $(\Delta x)^2 = (\Delta p)^2 = 1$. A textbook using $x = (a + a^\dagger)/\sqrt{2}$ has vacuum variance $1/2$; no physics changes, only the scaling.

4.2 The meaning of the phase-space plane

For a coherent state $|\alpha\rangle$,

$$\langle x\rangle = 2\,\mathrm{Re}\,\alpha, \quad \langle p\rangle = 2\,\mathrm{Im}\,\alpha, \quad (\Delta x)^2 = (\Delta p)^2 = 1.$$

Thus a coherent state looks like a vacuum uncertainty circle displaced from the origin. Its centre identifies a classical-like field amplitude; its width encodes unavoidable quantum fluctuations.



4.3 Rotated quadratures

For a local-oscillator phase θ,

$$\boxed{X_\theta = ae^{-i\theta} + a^\dagger e^{i\theta} = x\cos\theta + p\sin\theta.}$$

Homodyne detection measures the probability distribution of X_θ. At $\theta = 0$ it measures x; at $\theta = \pi/2$ it measures p. Measuring distributions over many angles enables optical homodyne tomography. The phase-space distribution is reconstructed from different projections, rather than measured pointwise with arbitrarily sharp x, p simultaneously.

Worked example: Rotate the local oscillator

Take $\alpha = 1 + i$. Then $\langle x \rangle = \langle p \rangle = 2$. At $\theta = \pi/4$,

$$\langle X_{\pi/4} \rangle = 2\cos(\pi/4) + 2\sin(\pi/4) = 2\sqrt{2}.$$

Its variance remains 1, since coherent vacuum noise is equally distributed in all directions.

4.4 Three complementary quasiprobabilities

The Wigner function $W(x, p)$ represents a quantum state in phase space. Its marginals give measurable quadrature probabilities, but some quantum states have negative Wigner regions. The Husimi function is

$$Q(\alpha) = \frac{1}{\pi} \langle \alpha | \rho | \alpha \rangle ,$$

which is always nonnegative and is smoother than the Wigner function. The Glauber–Sudarshan representation is

$$\rho = \int P(\alpha) |\alpha\rangle \langle \alpha| \, d^2\alpha.$$

Here P can be singular or nonpositive: it is *not always* a classical probability distribution. All three describe the same ρ, but highlight different physical properties.

4.5 Gaussian states and covariance

For $R = (x, p)^T$, define

$$d_i = \langle R_i \rangle, \qquad V_{ij} = \tfrac{1}{2} \langle \Delta R_i \Delta R_j + \Delta R_j \Delta R_i \rangle.$$

A Gaussian state is completely specified by d, V. Coherent and squeezed Gaussian states have Gaussian Wigner functions. A lossless linear transformation $R' = SR$ acts as $d' = Sd$, $V' = SVS^T$. An open-system Gaussian channel generally adds noise: $V' = SVS^T + N$.

Concept check

Phase space is useful for *states of modes*. Quadrature x is not position of a photon and p is not its momentum. They are conjugate field-amplitude variables.

5 Squeezing, coherence and photon correlations

Course sources: W3L2, W3L3; W6L3; Assignment 1

5.1 What is actually being squeezed?

Squeezing changes *field quadrature uncertainties*. A vacuum state has equal variance 1 in x, p with our convention. For real r, a squeezing operator can be defined as

$$S(r) = \exp\left[\frac{r}{2}(a^2 - a^{\dagger 2})\right].$$

Its Bogoliubov action is

$$S^\dagger a S = a\cosh r - a^\dagger \sinh r.$$

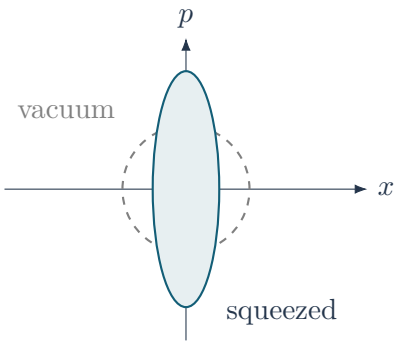
Consequently

$$S^\dagger x S = e^{-r}x, \qquad S^\dagger p S = e^{r}p,$$

and the squeezed vacuum $S(r)\,|0\rangle$ has

$$\boxed{(\Delta x)^2 = e^{-2r}, \quad (\Delta p)^2 = e^{2r}, \quad \Delta x\,\Delta p = 1.}$$

The uncertainty ellipse becomes narrow in one direction and wide in the other; the product does *not* violate Heisenberg.



5.2 How can vacuum contain squeezed photons?

The squeezed vacuum has $\langle a\rangle = 0$, but its mean photon number is

$$\boxed{\langle n\rangle = \sinh^2 r.}$$

It contains even-number superposition components produced by a parametric process. A *displaced squeezed* state has nonzero mean field as well as anisotropic noise. The order of $D(\alpha)$ and $S(r)$ matters if a calculation asks for the total number population, because the two operators do not generally commute.

5.3 Squeezing helps only the measured direction

If a signal shifts the x-mean by δx, the noise-limited sensitivity scales as $\delta x/\Delta x$. An x-squeezed probe reduces Δx by e^{-r} and improves this signal-to-noise ratio. But if one measures the p quadrature instead, the noise is *larger*. The optical phase of the local oscillator must therefore match the squeezing axis.

Worked example: How loss spoils squeezing

Take input x-quadrature variance $V_{\text{in}} = e^{-2r} = 1/4$, and a channel with power transmission $\eta = 0.8$. A vacuum environment enters through the unobserved beam-splitter port:

$$V_{\text{out}} = \eta V_{\text{in}} + (1-\eta)V_{\text{vac}} = 0.8(1/4) + 0.2(1) = \boxed{0.40}.$$

The output is still squeezed relative to vacuum ($V_{\text{out}} < 1$), but not as strongly as the input.

5.4 What does coherence measure?

First-order coherence measures field correlations across space or time:

$$G^{(1)}(t, t+\tau) = \langle a^\dagger(t) a(t+\tau) \rangle.$$

It is related to ordinary interference visibility and spectral linewidth. Second-order coherence measures correlated photon arrivals,

$$G^{(2)}(t, t+\tau) = \langle a^\dagger(t) a^\dagger(t+\tau) a(t+\tau) a(t) \rangle.$$

Normalised $g^{(2)}(0) > 1$ describes bunching; $g^{(2)}(0) < 1$ describes antibunching, which is nonclassical for a stationary single-mode field under conventional assumptions. An experiment can reveal coherence without being a direct measurement of a particular optical phase.

5.5 Pair production and heralding

A two-mode squeezed vacuum $\sum_n \lambda^n |n, n\rangle$ contains signal–idler number correlations. If the idler is unobserved, the signal looks thermal. If a detector heralds exactly one idler photon, the conditional signal is $|1\rangle$. This is possible because measurement changes the state of the *remaining* subsystem, even if the global evolution before detection was unitary.

Concept check

Squeezing concerns quadrature noise. Bunching concerns photon arrival statistics. First-order coherence concerns field correlations. These are related but not interchangeable ideas.

6 Photonic qubits and the Bloch sphere

Course sources: W4L1 and W4L2; Assignment 2

6.1 From an optical mode to a qubit

A qubit is a normalised superposition of two basis states, $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. For photons there are several useful physical encodings:

	$\lvert 0_L\rangle$	$\lvert 1_L\rangle$
polarisation	$\lvert H\rangle$	$\lvert V\rangle$
dual rail (path)	$\lvert 1,0\rangle$	$\lvert 0,1\rangle$
time bin	$\lvert\text{early}\rangle$	$\lvert\text{late}\rangle$

In dual rail, one photon is present across the two logical modes. The state $|0,0\rangle$ is an erasure/vacuum state outside the logical computational subspace, not logical zero.

6.2 What the Bloch sphere represents

Ignoring a global phase, every *pure* single-qubit state can be parameterised as

$$\boxed{|\psi\rangle = \cos(\theta/2)\,|0\rangle + e^{i\phi}\sin(\theta/2)\,|1\rangle\,.}$$

The Bloch vector is $\mathbf{r} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$. The poles represent $|0\rangle, |1\rangle$; the equator contains equal-amplitude superpositions with different relative phases. A mixed state has density matrix

$$\rho = \tfrac{1}{2}(I + \mathbf{r}\cdot\boldsymbol{\sigma}), \quad |\mathbf{r}| \le 1,$$

and lies inside the sphere.

6.3 Quantum gates are rotations

The Pauli matrices are

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Hadamard gate is $H = (X + Z)/\sqrt{2}$, with $H\,|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. A rotation generated by a Pauli operator is

$$R_j(\vartheta) = e^{-i\vartheta\sigma_j/2} = \cos(\vartheta/2)I - i\sin(\vartheta/2)\sigma_j.$$

On a path qubit, a beam splitter acts as a rotation and a relative optical phase acts as a Z-rotation. On a polarisation qubit, appropriately oriented wave plates implement the same mathematics. Physical optical hardware supplies a *representation* of the logical gate.

Worked example: A relative phase is observable

Start with $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. A phase shifter on logical $|1\rangle$ creates $(|0\rangle + e^{i\phi}\,|1\rangle)/\sqrt{2}$. Measuring immediately in the Z basis gives $1/2, 1/2$, regardless of ϕ. But applying a Hadamard before measurement gives

$$P(0) = \left|\frac{1 + e^{i\phi}}{2}\right|^2 = \boxed{\cos^2(\phi/2)}.$$

The interferometer converts a relative phase into a measurable number probability.

6.4 Entanglement is not classical correlation

Consider

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

It is a pure *joint* state that cannot be factored into independent states of Alice and Bob. Trace out Bob:

$$\rho_A = \mathrm{Tr}_B \left|\Phi^+\right\rangle \left\langle\Phi^+\right| = \tfrac{1}{2} I.$$

Alice's measurement is locally random, yet her result matches Bob's if both measure in the same Z basis. In the X basis, $\left|\Phi^+\right\rangle = (|{+}{+}\rangle + |{-}{-}\rangle)/\sqrt{2}$, so the correlations persist there too.

6.5 Partial trace is a practical calculation

For a two-qubit density operator expressed as $\rho = \sum_{ijkl} \rho_{ij,kl} \left|i,j\right\rangle \left\langle k,l\right|$,

$$\boxed{\mathrm{Tr}_B\, \rho = \sum_{i,k,j} \rho_{ij,kj} \left|i\right\rangle \left\langle k\right|.}$$

The two indices of the discarded subsystem must agree. This erases correlations not accessible to measurements on A alone.

Concept check

A pure entangled pair can have *mixed* single-photon reduced states. The Bloch sphere describes a qubit's local state; it does not by itself encode all joint correlations.

7 Quantum communication: teleportation, dense coding and QKD

Course sources: W4L2–W4L3; Assignment 2; 2025 Q4(e)

7.1 Three communication tasks: do not confuse them

Teleportation transfers an *unknown qubit state* using a shared entangled pair and two classical bits. Dense coding sends *two classical bits* using one transmitted qubit and one pre-shared Bell pair. Quantum key distribution (QKD) establishes shared secret classical bits and tests whether an eavesdropper could have learned them. The resource accounting is different in each task.

7.2 Teleportation as a calculation

Suppose Alice holds $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Alice and Bob also share $|\Phi^+\rangle_{AB} = (|00\rangle + |11\rangle)/\sqrt{2}$. Alice performs CNOT from the unknown qubit onto her half of the Bell pair, then a Hadamard on the unknown qubit. Rewriting in Alice's two-qubit computational basis gives

$$|\psi\rangle\left|\Phi^+\right\rangle_{AB} \longrightarrow \frac{1}{2}[|00\rangle_A|\psi\rangle_B + |01\rangle_A X|\psi\rangle_B + |10\rangle_A Z|\psi\rangle_B + |11\rangle_A XZ|\psi\rangle_B],$$

up to a harmless convention-dependent phase on the last branch. Alice measures her two qubits and sends the result (two bits). Bob applies the appropriate Pauli correction:

$$00 \to I, \qquad 01 \to X, \qquad 10 \to Z, \qquad 11 \to ZX$$

(with the order consistent with the state branch). The outcome is always Bob's $|\psi\rangle$. The original input has been consumed, so this does not clone the state. Bob cannot recover $|\psi\rangle$ before Alice's classical message arrives.

7.3 Dense coding uses four Bell messages

Starting from $|\Phi^+\rangle$, Alice can apply

$$I \to \Phi^+, \quad X \to \Psi^+, \quad Z \to \Phi^-, \quad XZ \to \Psi^-$$

(up to global phase). Bob would ideally discriminate all four orthogonal Bell states after receiving Alice's photon. The standard passive beam-splitter and number-detection analyser distinguishes $\Psi^\pm$ through their port/polarisation patterns, but not Φ^+ from Φ^-. Thus the ordinary no-ancilla linear-optical Bell measurement is not deterministic; its unambiguous success probability for equiprobable Bell states is $1/2$.

7.4 BB84: prepare, measure, sift, test

BB84 uses two conjugate bases,

$$Z : \{|H\rangle, |V\rangle\}, \qquad X : \{|D\rangle, |A\rangle\}, \quad |D/A\rangle = \frac{|H\rangle \pm |V\rangle}{\sqrt{2}}.$$

Alice randomly chooses a basis and bit; Bob randomly chooses a measurement basis. After transmission they announce *bases*, not retained secret bit values, and keep measurements where their bases matched. Independent uniform bases match with probability $\boxed{1/2}$. With channel and detector efficiencies η_{ch}, η_d, the expected sifted number out of N ideal single-photon pulses is

$$\boxed{\mathbb{E}[N_{\text{sift}}] = \frac{N}{2}\eta_{\text{ch}}\eta_d},$$

before testing, reconciliation and privacy amplification.

Worked example: Why intercept–resend creates 25% errors

Condition on a sifted event: Bob used Alice's basis. An intercept–resend Eve chooses that basis with probability 1/2, making no error. If Eve used the wrong basis (probability 1/2), Bob's subsequently recovered bit is wrong with probability 1/2. Therefore

$$\boxed{\text{QBER} = \tfrac{1}{2}(0) + \tfrac{1}{2}(\tfrac{1}{2}) = \tfrac{1}{4}.}$$

If $m = 20$ independent sifted test bits are revealed, the probability of noticing at least one error in this simple model is $1 - (3/4)^{20} \approx 0.9968$. These public test bits are discarded.

7.5 What makes a practical key secure?

Alice and Bob estimate a quantum bit error rate, perform error correction and then privacy amplification. Revealed parity information and imperfect sources affect the final key rate. Security requires an *authenticated* classical discussion channel as well as quantum transmission. Multi-photon weak-laser pulses are not identical to ideal single-photon states; decoy-state methods are one way to address source imperfections in implementations.

Concept check

Teleportation is not instantaneous transport of an object. Dense coding needs prior entanglement. BB84 detects disturbance through *incompatible measurements*, not through a secret choice of quantum bases.

8 Noise, loss and quantum channels

Course sources: W5L1; Week 5 tutorial

8.1 Why a subsystem does not evolve unitarily

For an isolated quantum system, $\rho' = U\rho U^\dagger$. An optical mode in the laboratory interacts with unobserved environmental degrees of freedom: lost photons, thermally excited modes, unwanted phase fluctuations and detectors. If the environment starts in ρ_E,

$$\boxed{\rho'_S = \mathrm{Tr}_E[U_{SE}(\rho_S \otimes \rho_E)U_{SE}^\dagger].}$$

The joint evolution is unitary but the reduced system evolution generally is not. Tracing out the environment is an *average over unobserved information*, not a measurement that selects a particular outcome.

8.2 Loss as a beam splitter with vacuum

Introduce an initially empty environment mode e:

$$a_{\mathrm{out}} = \sqrt{\eta}\, a_{\mathrm{in}} + \sqrt{1-\eta}\, e_{\mathrm{in}}.$$

This is the physically correct quantum loss model; replacing a by $\sqrt{\eta}a$ alone would give $[a_{\mathrm{out}}, a_{\mathrm{out}}^\dagger] = \eta \neq 1$. The environment operator supplies the vacuum noise that preserves the commutator. For a single photon,

$$|1,0\rangle_{S,E} \longrightarrow \sqrt{\eta}\,|1,0\rangle + \sqrt{1-\eta}\,|0,1\rangle\,.$$

Tracing over the environment gives

$$\boxed{\rho'_S = \eta\,|1\rangle\langle 1| + (1-\eta)\,|0\rangle\langle 0|\,.}$$

The probability of having a photon is η; the environment is responsible for the mixture.

8.3 Kraus operators: a system-only description

The same channel can be written

$$\boxed{\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I.}$$

For a single-rail state supported on $\{|0\rangle, |1\rangle\}$, loss has

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{\eta} \end{pmatrix}, \qquad K_1 = \begin{pmatrix} 0 & \sqrt{1-\eta} \\ 0 & 0 \end{pmatrix}.$$

The K_0 branch contains no environment excitation, while K_1 represents losing the photon. These are *not* two alternative deterministic maps; they form one trace-preserving channel when summed.

Worked example: A superposition passes through loss

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

$$\rho' = \sum_k K_k|\psi\rangle\langle\psi|K_k^\dagger = \boxed{\begin{pmatrix} |\alpha|^2 + (1-\eta)|\beta|^2 & \sqrt{\eta}\,\alpha\beta^* \\ \sqrt{\eta}\,\alpha^*\beta & \eta|\beta|^2 \end{pmatrix}}.$$

Check the trace: it is 1. The excited population is suppressed by η, but vacuum–one-photon coherence by $\sqrt{\eta}$. For $\eta = 1$ the original pure state is recovered. For $\eta = 0$ the output is vacuum.

8.4 Phase damping is different from photon loss

Phase noise can suppress coherences without changing photon-number probabilities. For a number basis, a random phase $e^{i\phi\hat{n}}$ has no effect on diagonal populations but averages away off-diagonal terms such as $|0\rangle\langle 1|$. A simple dephasing channel acts on a qubit as

$$\rho_{01} \mapsto \lambda\rho_{01}, \quad \rho_{00} \mapsto \rho_{00}, \quad \rho_{11} \mapsto \rho_{11}, \qquad 0 \le \lambda \le 1.$$

Hence direct photon counting can miss phase noise that is obvious in homodyne detection.

8.5 When measurement becomes conditioning

If an environmental or ancilla outcome k is actually observed and kept, use

$$\tilde{\rho}_k = K_k \rho K_k^\dagger, \quad p_k = \operatorname{Tr}\tilde{\rho}_k, \quad \boxed{\rho_k = \tilde{\rho}_k / p_k}.$$

This is a trace-decreasing *conditional* operation. Summing over all outcomes instead recovers the unconditional channel. For conditional photon subtraction, a weak beam splitter followed by a one-photon detector click implements an operator approximately proportional to a. It weights higher Fock components by $\sqrt{n}$, not simply removing one photon deterministically.

Concept check

Ask which information is available. If the environment is ignored, trace it out or sum Kraus branches. If a detector outcome is heralded, *select and renormalise* that branch, retaining its separate success probability.

9 Damping, Langevin noise and master equations

Course sources: W5L2; 2024 Q4; 2025 Q4

9.1 Two ways of describing the same physics

The Heisenberg picture evolves operators; the Schrödinger picture evolves density matrices. For a closed system,

$$\dot{O} = \frac{i}{\hbar}[H, O], \qquad \dot{\rho} = -\frac{i}{\hbar}[H, \rho].$$

An open system also has damping and fluctuations, because it couples to many unobserved bath modes. The Markov approximation replaces bath memory by effectively instantaneous noise; it is useful when bath correlations decay quickly compared with the system dynamics.

9.2 Where the quantum Langevin equation comes from

A cavity mode a coupled weakly to a continuum of oscillators b_ω has a Hamiltonian of the schematic form

$$H/\hbar = \omega_c a^\dagger a + \int \omega b_\omega^\dagger b_\omega d\omega + i \int \kappa(\omega)(a^\dagger b_\omega - b_\omega^\dagger a) d\omega.$$

One first solves the bath equation for $b_\omega(t)$; substituting it back into the cavity equation produces a *damping* term and an *input noise* term. In a frame rotating at the cavity resonance,

$$\boxed{\dot{a} = -\frac{\gamma}{2} a + \sqrt{\gamma}\, a_{\text{in}}(t)}$$

for an undriven cavity with one damping port. The noise operator is not an optional classical disturbance: it ensures that operator commutators remain physically valid.

9.3 Fourier solving is usually easiest

Define $\tilde{a}(\omega) = \int_{-\infty}^{\infty} a(t) e^{-i\omega t}\, dt$, so $\widetilde{\dot{a}} = i\omega \tilde{a}$ when boundary terms vanish. Then

$$(i\omega + \gamma/2)\tilde{a} = \sqrt{\gamma}\, \tilde{a}_{\text{in}}, \qquad \boxed{\tilde{a} = \frac{\sqrt{\gamma}}{\gamma/2 + i\omega} \tilde{a}_{\text{in}}}.$$

The denominator is the cavity susceptibility. Near zero detuning it has a large response; far from resonance the response is smaller. A consistent Fourier convention is essential because another sign convention replaces $i\omega$ by $-i\omega$.

9.4 Input–output theory

With one common convention,

$$a_{\text{out}} = \sqrt{\gamma}\, a - a_{\text{in}}.$$

Eliminating the intracavity operator gives a frequency-dependent relation between input and output fields. For a two-port cavity with $\gamma = \gamma_1 + \gamma_2$,

$$\dot{a} = -\frac{\gamma}{2} a + \sqrt{\gamma_1} a_{\text{in},1} + \sqrt{\gamma_2} a_{\text{in},2},$$

$$\boxed{\tilde{a}_{\text{out},1} = \frac{\gamma_1 - \gamma/2 - i\omega}{\gamma/2 + i\omega} \tilde{a}_{\text{in},1} + \frac{\sqrt{\gamma_1 \gamma_2}}{\gamma/2 + i\omega} \tilde{a}_{\text{in},2}.}$$

At $\omega = 0$ and $\gamma_1 = \gamma_2$ the coherent reflection coefficient vanishes. But the second input's vacuum fluctuations must remain in the *field operator*.

9.5 The Lindblad master equation

After tracing over a Markovian bath, a general trace-preserving reduced evolution is

$$\boxed{\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + \sum_j \gamma_j \mathcal{D}[L_j]\rho},$$

where

$$\mathcal{D}[L]\rho = L\rho L^\dagger - \frac{1}{2}(L^\dagger L\rho + \rho L^\dagger L).$$

A cavity losing photons into vacuum has $L = a$, while a two-level atom decaying spontaneously has $L = \sigma_-$. Pure dephasing can be modelled with a Hermitian jump operator related to σ_z. The jump operator tells you *which physical transition the environment causes*.

Worked example: Photon number decays exponentially

For zero-temperature cavity loss, $\dot{\rho} = \gamma\mathcal{D}[a]\rho$. Use $\frac{d}{dt}\langle n\rangle = \mathrm{Tr}(n\dot{\rho})$ and $[n, a] = -a$. After rearrangement,

$$\frac{d}{dt}\langle n\rangle = -\gamma\langle n\rangle.$$

Hence

$$\boxed{\langle n(t)\rangle = \langle n(0)\rangle e^{-\gamma t}}.$$

The field amplitude decays at half that exponent $(\gamma/2)$, explaining the factor 2 in the Langevin equation.

9.6 Parametric squeezing in a cavity

For interaction $H_I = (i\hbar\chi/2)(a^2 - a^{\dagger 2})$, the Heisenberg commutator contributes $-\chi a^\dagger$ to $\dot{a}$. Define $X_+ = a + a^\dagger$, giving

$$\dot{X}_+ = -(\gamma/2 + \chi)X_+ + \sqrt{\gamma}X_{+,\mathrm{in}}.$$

The output noise transfer factor leads to

$$\boxed{V_+(\omega) = \frac{(\gamma/2 - \chi)^2 + \omega^2}{(\gamma/2 + \chi)^2 + \omega^2}}$$

for unit vacuum input spectrum and $0 < \chi < \gamma/2$. The *variance* comes from field-fluctuation correlations, not the square of the mean field.

Concept check

For a Langevin exam question: identify the Hamiltonian and port rates, calculate the commutator, Fourier transform, apply the exact input–output sign convention, then evaluate the noise spectrum.

10 Conditional optics: how measurement becomes a gate

Course sources: W5L3; Week 6 tutorial; Assignment 2

10.1 A surprising capability of linear optics

Beam splitters and phase shifters implement linear unitary transformations of optical modes. They cannot directly create a deterministic interaction between two photons. But combining these optical elements with ancilla photons and *postselected* photodetection can implement an effective nonlinear operation on a conditional subensemble. The cost is that some runs must be rejected.

10.2 One outcome, one Kraus operator

If the joint state of signal and ancilla is $U(|\psi\rangle \otimes |\text{anc}\rangle)$ and the ancilla detector yields k, the signal's unnormalised output is

$$\left|\tilde{\psi}_k\right\rangle = {}_{\text{anc}}\langle k|\, U(|\psi\rangle \otimes |\text{anc}\rangle),$$

or $\tilde{\rho}_k = K_k \rho K_k^\dagger$. The probability of this outcome is $p_k = \operatorname{Tr} \tilde{\rho}_k$; $\rho_k = \tilde{\rho}_k / p_k$ is the normalised state for successful trials. Do *not* normalise before calculating the success probability.

10.3 Photon subtraction

Consider a signal $|\psi\rangle = \sum_n c_n |n\rangle$ mixed with vacuum at a weakly reflecting beam splitter of transmissivity η. Conditioning on one photon in the reflected output produces

$$\boxed{K_{\text{click}} |n\rangle \propto \sqrt{n}\,\sqrt{1-\eta}\,\eta^{(n-1)/2}\,|n-1\rangle\,.}$$

The factor $\sqrt{n}$ is the annihilation-operator matrix element. The factor $\eta^{(n-1)/2}$ accounts for the photons retained in the signal. For η near 1, the operation approximates $a\,|\psi\rangle$, up to an overall success amplitude. It *does not* remove a photon deterministically from every trial.

Worked example: Distinguishing nonorthogonal states probabilistically

Take the two input states

$$|\psi_\pm\rangle = \frac{2\,|0\rangle \pm |1\rangle}{\sqrt{5}}, \qquad \langle\psi_+|\psi_-\rangle = 3/5.$$

They cannot be perfectly distinguished by a deterministic measurement. Use a successful filter

$$K_s = \tfrac{1}{2}\,|0\rangle\langle 0| + |1\rangle\langle 1|\,.$$

The unnormalised output is $(|0\rangle \pm |1\rangle)/\sqrt{5}$ with probability $p_s = 2/5$. After normalising,

$$\boxed{|\phi_\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}}.$$

These are *orthogonal*, so a Hadamard-basis measurement now identifies the sign, conditional on success. The failure probability is 3/5; the identical failure output contains no information about the sign. Probabilistic filtering does not violate no-cloning or deterministic distinguishability.

10.4 Heralding and gate resources

Conditional sign gates can transform selected Fock coefficients with relative phases, for example $\alpha\,|0\rangle + \beta\,|1\rangle + \gamma\,|2\rangle$ into $\alpha\,|0\rangle + \beta\,|1\rangle - \gamma\,|2\rangle$ in a successful ancilla detection branch. Such number-dependent relative phases are effectively nonlinear operations, even though the optical elements before

measurement were linear. This idea is used in measurement-induced photonic computation, with ancilla states, feed-forward and suitable encoding.

10.5 Entanglement distillation

Two noisy Bell pairs can be processed with bilateral gates and local measurements to keep one pair only when parity checks succeed. For the simple ensemble

$$\rho = p\left|\Phi^+\right\rangle\left\langle\Phi^+\right| + (1-p)\left|\Psi^+\right\rangle\left\langle\Psi^+\right|,$$

a recurrence protocol keeps matching error labels with

$$p_{\text{success}} = p^2 + (1-p)^2, \qquad \boxed{p' = \frac{p^2}{p^2 + (1-p)^2}}.$$

For $p = 0.8$, the retained fidelity rises to $p' \approx 0.9412$, but the pair is obtained with probability 0.68 from two initial pairs. Entanglement quality improves at the expense of throughput.

Concept check

Always report *both* the conditional output state and its success probability. Distinguish a trace-preserving quantum channel from a selected, trace-decreasing instrument outcome.

11 One atom, one photon: Jaynes–Cummings physics

Course sources: W6L1; W6L2; Week 6 tutorial

11.1 Why a two-level atom is a useful model

An atom has many levels, but when a narrow-band cavity is tuned close to a single transition $|g\rangle \leftrightarrow |e\rangle$, the other levels may be neglected to a good approximation. The atom then behaves like a spin-1/2 system with

$$\sigma_+ = |e\rangle\langle g|, \quad \sigma_- = |g\rangle\langle e|, \quad \sigma_z = |e\rangle\langle e| - |g\rangle\langle g|.$$

One photon of frequency ω_c has the right energy to excite the atom if $\omega_c \approx \omega_a$. The dipole interaction $H_{\text{int}} = -\mathbf{d}\cdot\mathbf{E}$ contains products of photon and atomic raising/lowering operators.

11.2 Rotating-wave approximation

Terms $a\sigma_+$ and $a^\dagger\sigma_-$ exchange excitation between atom and field. The counter-rotating combinations $a\sigma_-$ and $a^\dagger\sigma_+$ change excitation number by two and oscillate rapidly near resonance. If $g \ll \omega_a, \omega_c$, the rotating-wave approximation retains the slowly varying terms:

$$\boxed{H_{\text{JC}} = \hbar\omega_c a^\dagger a + \hbar\omega_a \sigma_+\sigma_- + \hbar g(a\sigma_+ + a^\dagger\sigma_-).}$$

The excitation operator $N = a^\dagger a + \sigma_+\sigma_-$ commutes with H_{JC}: excitations move, but their total number is conserved.

11.3 Reduce the problem to a two-state block

For a fixed total excitation, the coupled states are $|e,n\rangle$ and $|g,n+1\rangle$. After subtracting a common energy, the Hamiltonian block is

$$\frac{H_n}{\hbar} = \frac{1}{2}\begin{pmatrix} \Delta & 2g\sqrt{n+1} \\ 2g\sqrt{n+1} & -\Delta \end{pmatrix}, \quad \Delta = \omega_a - \omega_c.$$

Its eigenfrequency separation is

$$\boxed{\Omega_n = \sqrt{\Delta^2 + 4g^2(n+1)}.}$$

The factor $\sqrt{n+1}$ comes from $a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle$. It is a directly observable signature of a quantised cavity field.

Worked example: Vacuum Rabi oscillation

Begin in $|e,0\rangle$ at resonance ($\Delta = 0$). The interaction block is $\hbar g\sigma_x$, so

$$|\psi(t)\rangle = \cos(gt)\,|e,0\rangle - i\sin(gt)\,|g,1\rangle.$$

Thus

$$\boxed{P_e(t) = \cos^2(gt), \quad P_{g,1}(t) = \sin^2(gt).}$$

At $t = \pi/(2g)$ the atom has emitted one photon into the cavity. At a full period $t = \pi/g$, the excitation has returned to the atom.

11.4 From one atom to many: Tavis–Cummings

For N identical emitters with the same coupling phase,

$$H_I = \hbar g(aJ_+ + a^\dagger J_-), \quad J_\pm = \sum_{i=1}^{N} \sigma_\pm^{(i)}.$$

For two atoms with one excitation, define

$$|S\rangle = \frac{|eg\rangle + |ge\rangle}{\sqrt{2}}, \quad |A\rangle = \frac{|eg\rangle - |ge\rangle}{\sqrt{2}}.$$

Because

$$J_- |S\rangle = \sqrt{2}\,|gg\rangle\,, \qquad J_- |A\rangle = 0,$$

the symmetric state is *bright*, with coupling $\sqrt{2}g$; the antisymmetric state is *dark* to the common cavity mode.

11.5 Dicke states and superradiant enhancement

For N symmetric spins, collective angular-momentum labels $J = N/2, m$ describe a Dicke ladder. The lowering operator has matrix element

$$J_- |J, m\rangle = \sqrt{(J+m)(J-m+1)}\,|J, m-1\rangle\,.$$

If all atoms couple to a *common* radiation reservoir under appropriate conditions, these factors enhance collective spontaneous emission rates. The coherent cavity-exchange enhancement $\sqrt{N}\,g$ and cooperative radiative superradiance are related but not identical phenomena: the reservoir and initial state also matter.

Worked example: Two-atom conservation check

Start with $|eg, 0\rangle$. It equals $(|S, 0\rangle + |A, 0\rangle)/\sqrt{2}$. The bright half oscillates with $\sqrt{2}g$, while the dark half stays. Writing $\vartheta = \sqrt{2}gt$,

$$|\psi(t)\rangle = \frac{\cos\vartheta\,|S, 0\rangle + |A, 0\rangle - i\sin\vartheta\,|gg, 1\rangle}{\sqrt{2}}.$$

The mean cavity photon number is $\langle n\rangle = \frac{1}{2}\sin^2\vartheta$. The missing probability remains as atomic excitation, preserving $\langle N\rangle = 1$.

Concept check

To solve cavity-QED problems, identify the conserved excitation number first. That reduces an infinite-dimensional Hamiltonian to independent 2×2 or small collective subspaces.

12 Quantum metrology and interferometers

Course sources: W6L3; 2023 Q3; gravitational-wave application

12.1 A parameter is not the same as an observation

Suppose the quantity we wish to know is an optical phase ϕ. The state depends on ϕ, a measurement yields random data x, and an estimator converts those data into a guess $\tilde{\phi}$. Even an ideal apparatus cannot generally extract an arbitrarily precise value from one quantum measurement. Repeated trials reduce statistical uncertainty.

12.2 A Mach–Zehnder turns phase into intensity

A balanced beam splitter divides an input into two paths. One path accumulates a relative phase; a second splitter recombines the alternatives. For one photon entering a chosen input port,

$$\boxed{P_1(\phi) = \cos^2(\phi/2), \quad P_2(\phi) = \sin^2(\phi/2).}$$

These probabilities oscillate with phase even though the photon count before recombination did not reveal which relative phase was present.



12.3 Fisher information measures sensitivity

The classical Fisher information for a set of outcomes x is

$$\boxed{F(\phi) = \sum_x p(x|\phi)[\partial_\phi \ln p(x|\phi)]^2 = \sum_x \frac{(\partial_\phi p_x)^2}{p_x}.}$$

A large F means a small phase change strongly changes the outcome distribution. For M independent measurements, the classical Cramér–Rao bound is

$$\mathrm{Var}(\tilde{\phi}) \geq \frac{1}{MF(\phi)}$$

for unbiased estimators under the usual regularity assumptions.

Worked example: Information in a single-photon interferometer

For $p_1 = \cos^2(\phi/2)$, $p_2 = \sin^2(\phi/2)$, differentiate:

$$\partial_\phi p_1 = -\tfrac{1}{2}\sin\phi, \qquad \partial_\phi p_2 = +\tfrac{1}{2}\sin\phi.$$

Hence

$$F = \frac{\sin^2\phi}{4}\left[\frac{1}{\cos^2(\phi/2)} + \frac{1}{\sin^2(\phi/2)}\right] = \boxed{1},$$

with the same continuous limit at endpoints. The phase standard deviation cannot be smaller than $1/\sqrt{M}$ in this ideal single-photon measurement model.

12.4 Quantum Fisher information

The quantum Fisher information F_Q is the largest information obtainable by optimising over possible measurements. For a pure probe encoded through $|\psi_\phi\rangle = e^{-i\phi G}|\psi_0\rangle$,

$$\boxed{F_Q = 4\,\mathrm{Var}_{\psi_0}(G).}$$

An optical phase generated by photon number has $G = n$. For a coherent state $\mathrm{Var}(n) = \bar{n}$, so $F_Q = 4\bar{n}$, giving the characteristic shot-noise scaling $\Delta\phi \propto 1/\sqrt{\bar{n}}$. Different conventions for whether the parameter is the phase on one arm or the relative arm phase alter constant factors in the generator, but not this scaling.

12.5 How squeezing can help, and when it cannot

A squeezed input can improve sensitivity when its quiet quadrature matches the noise relevant to the measurement. In a gravitational-wave interferometer, squeezed vacuum is often injected into an otherwise unused port to reduce measured quantum fluctuations. However, *one squeezed-coherent input plus vacuum in the other port*, measured by simple output-number difference at a balanced operating phase, can still give variance equal to total input mean photon number. That specific configuration does not automatically beat shot-noise scaling at equal energy.

Concept check

When asked for precision, identify: the encoded parameter, generator G, actual measurement outcomes, photon resource, repetitions and operating phase. QFI is an *optimised bound*; a particular detector may not attain it.

13 Hydrogenic atoms and radial wavefunctions

Course sources: Week 8; 2023 Q2; Assignment 3 Q1; Assignment 4 Q3

13.1 The one-electron Schrödinger equation

For an electron of reduced mass μ around a point nucleus of charge $+Ze$,

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 - \frac{Ze^2}{4\pi\epsilon_0 r}\right]\psi(\mathbf{r}) = E\psi(\mathbf{r}).$$

The Coulomb potential depends only on the radius r, so angular momentum is conserved. Separate variables into

$$\boxed{\psi_{nlm}(\mathbf{r}) = R_{nl}(r)Y_{lm}(\theta,\phi).}$$

The spherical harmonics are eigenfunctions of L^2, L_z with

$$L^2 Y_{lm} = \hbar^2 l(l+1)Y_{lm}, \qquad L_z Y_{lm} = \hbar m Y_{lm}.$$

The labels have allowed values $n = 1, 2, \ldots$, $l = 0, \ldots, n-1$, $m = -l, \ldots, l$.

13.2 The centrifugal barrier

Define $u(r) = rR_{nl}(r)$. Then the radial Schrödinger equation becomes

$$\boxed{-\frac{\hbar^2}{2\mu}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2 l(l+1)}{2\mu r^2}\right]u = Eu.}$$

The term proportional to $1/r^2$ is the effective centrifugal barrier. For $l > 0$, the radial wavefunction is suppressed near the nucleus; s-states with $l = 0$ penetrate it much more strongly. The normalisation can be written

$$\int_0^\infty |R_{nl}|^2 r^2 dr = \int_0^\infty |u_{nl}|^2 dr = 1.$$

Switching from R to u changes the *measure* in every radial matrix element.

13.3 Why hydrogen energies depend only on n

The nonrelativistic Coulomb levels are

$$\boxed{E_n = -\frac{\mu Z^2 e^4}{2(4\pi\epsilon_0)^2\hbar^2 n^2} \approx -13.6\,\text{eV}\,\frac{\mu}{m_e}\frac{Z^2}{n^2}.}$$

The n-only degeneracy is a special symmetry of the pure Coulomb potential. Screening in a many-electron atom, relativity, magnetic fields and finite nuclear size lift different parts of this degeneracy.

Worked example: Most probable radius is not mean radius

For hydrogen $1s$, $\psi_{1s} = e^{-r/a_0}/\sqrt{\pi a_0^3}$. The probability in a spherical shell between r and $r + dr$ is

$$P(r)dr = 4\pi r^2|\psi|^2 dr = \frac{4r^2}{a_0^3}e^{-2r/a_0}dr.$$

Differentiate $\ln P(r)$: $2/r - 2/a_0 = 0$, so $\boxed{r_{\text{most probable}} = a_0}$. But

$$\langle r\rangle = \frac{4}{a_0^3}\int_0^\infty r^3 e^{-2r/a_0}dr = \boxed{3a_0/2}.$$

The different answers are not contradictory: one describes the peak of a distribution; the other its average.

13.4 Finite nuclear size is a perturbation

A nucleus has a finite radius and a spread-out charge distribution. The point Coulomb potential $-Z/r$ is too singular near the origin. For a uniform nuclear sphere of radius R in atomic units,

$$V_{\text{finite}}(r) = -\frac{Z}{2R}\left(3 - \frac{r^2}{R^2}\right) \quad (r < R).$$

The perturbation is

$$\Delta V = V_{\text{finite}} - V_{\text{point}} = Z\left(\frac{1}{r} - \frac{3}{2R} + \frac{r^2}{2R^3}\right).$$

First-order perturbation theory gives

$$\boxed{\Delta E_{nl} = \int_0^R |R_{nl}(r)|^2 r^2 \Delta V(r)\, dr.}$$

If R is small and the s-state wavefunction is approximately constant inside the nucleus, one obtains

$$\boxed{\Delta E_{ns} \simeq \frac{2Z^4R^2}{5n^3} \quad \text{(Hartree)}.}$$

The positive shift makes the state less strongly bound. It is particularly large for orbitals with nonzero density at the origin.

13.5 How numerical radial solutions work

For a screened or model potential $V(r)$, the radial equation is often discretised. At grid spacing h,

$$u''(r_i) \approx \frac{u(r_{i-1}) - 2u(r_i) + u(r_{i+1})}{h^2}.$$

The Hamiltonian becomes a sparse tridiagonal matrix. Diagonalising it yields energies and radial eigenvectors. Check normalisation, radial nodes and stability as h decreases and the outer boundary moves outward. The sodium lifetime assignment uses exactly this kind of numerical workflow.

Concept check

Hydrogenic n, l, m label wavefunctions, not miniature classical orbits. For a radial integral, always decide whether you are using $R(r)$ or $u(r) = rR(r)$ before writing the measure.

14 Many-electron atoms, configurations and term symbols

Course sources: Week 8; Assignment 3 Q2; Assignment 4 Q1

14.1 Why the hydrogen formula stops working

Two electrons repel each other. The many-electron Hamiltonian has the form

$$\boxed{H = \sum_i \left[\frac{p_i^2}{2m_e} - \frac{Ze^2}{4\pi\epsilon_0 r_i}\right] + \sum_{i<j} \frac{e^2}{4\pi\epsilon_0 r_{ij}}.}$$

The last term depends on *two* electron positions, so we cannot solve each electron independently as if the other electrons were absent. An approximate central potential describes the nucleus screened by the electron cloud. The one-electron orbitals are labelled n, l, but their energies are no longer functions of n alone.

14.2 Pauli exclusion and antisymmetric states

Electrons have spin $s = 1/2$ and are identical fermions. Interchanging two electron labels must reverse the *total* wavefunction:

$$\Psi(1,2) = -\Psi(2,1).$$

A two-electron Slater determinant automatically enforces this:

$$\Psi_{ab}(1,2) = \frac{1}{\sqrt{2}}[a(1)b(2) - b(1)a(2)].$$

If $a = b$, the determinant vanishes: two electrons cannot occupy the same spin-orbital.

14.3 Direct repulsion and exchange

Evaluate Coulomb repulsion $v_{12} = e^2/(4\pi\epsilon_0 r_{12})$ in a Slater determinant. Expanding the bra and ket yields

$$\boxed{\langle v_{12}\rangle = J_{ab} - K_{ab}},$$

where

$$J_{ab} = \iint \frac{|a(1)|^2|b(2)|^2}{r_{12}}\, d1\, d2,$$

$$K_{ab} = \iint \frac{a^*(1)b(1)b^*(2)a(2)}{r_{12}}\, d1\, d2$$

in atomic units. The direct term resembles ordinary Coulomb repulsion between densities. The exchange term arises from the antisymmetric *wavefunction*. It is not another classical force; for orthogonal spin labels it vanishes. Hartree–Fock finds orbitals self-consistently so that direct and exchange contributions enter the mean-field equations.

14.4 How spectroscopic term notation works

For light atoms with reasonably good LS (Russell–Saunders) coupling, first add the individual orbital momenta to L, individual electron spins to S, then couple L, S to total electronic J. Write

$$\boxed{{}^{2S+1}L_J}, \qquad L = 0, 1, 2, 3, 4 \leftrightarrow S, P, D, F, G.$$

For example, ${}^3F_2$ means total spin $S = 1$ (triplet), total orbital angular momentum $L = 3$ and total electronic $J = 2$. The allowed J values are $J = |L - S|, |L - S| + 1, \ldots, L + S$.

14.5 Equivalent electrons require extra care

Two d electrons each have $l = 2$ and two spin projections. There are ten distinct spin-orbitals, hence

$$\binom{10}{2} = 45$$

antisymmetric configurations. Without Pauli restrictions, angular-momentum addition would suggest $L = 0, 1, 2, 3, 4$, $S = 0, 1$. But the orbital-coupled wavefunction has exchange phase $(-1)^{2l-L} = (-1)^L$. The spin singlet $S = 0$ is antisymmetric, while triplet $S = 1$ is symmetric. Thus only *even* L, $S = 0$ or *odd* L, $S = 1$ survive:

$$\boxed{{}^1S_0,\ {}^1D_2,\ {}^1G_4,\quad {}^3P_{0,1,2},\ {}^3F_{2,3,4}.}$$

Worked example: Check that every d^2 state is counted

A term with total J has $2J + 1$ magnetic substates. Sum:

$$1 + 5 + 9 + (1 + 3 + 5) + (5 + 7 + 9) = \boxed{45}.$$

The count agrees with the Pauli spin-orbital count $\binom{10}{2}$. Without this check we could easily include forbidden LS terms or miss part of an allowed multiplet.

14.6 Hund's rules and why they are only trends

Within their usual domain, Hund's rules favour (1) largest total S, then (2) largest L for that multiplicity, then (3) smallest J in a less-than-half-filled subshell. For d^2, this predicts the ground level ${}^3F_2$. However, these rules do *not* unambiguously order every excited term of a real multi-electron atom. Titanium's neutral $3d^2 4s^2$ configuration demonstrates the broad agreement and exceptions when compared with NIST experimental level data.

Concept check

The notation ${}^{2S+1}L_J$ is a coupling scheme, not a claim that electrons orbit independently. First impose antisymmetry, then enumerate LS terms, attach J, and check the total degeneracy.

15 Adding angular momenta, Clebsch–Gordan and 3-j symbols

Course sources: Week 8; 2025 Q1; Atomic Assignment 4 Q1

15.1 What is being added?

An electron has orbital angular momentum $\mathbf{L}$ and spin $\mathbf{S}$; an atom may have nuclear angular momentum $\mathbf{I}$. The sum of two angular momentum vectors is again an angular momentum operator: $\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2$. But a simultaneous eigenstate of J_1^2, J_2^2, J^2, J_z is not generally a product of separate J_{1z} and J_{2z} eigenstates.

Angular momentum eigenvalues are

$$J^2 |jm\rangle = \hbar^2 j(j+1) |jm\rangle , \qquad J_z |jm\rangle = \hbar m |jm\rangle .$$

For j_1, j_2, the allowed total j values are

$$\boxed{j = |j_1 - j_2|,\ |j_1 - j_2| + 1, \ldots, j_1 + j_2.}$$

The total projection is $m = m_1 + m_2$. That is the *angular-momentum addition rule.*

15.2 What does a Clebsch–Gordan coefficient say?

In the product basis, $|j_1 m_1\rangle |j_2 m_2\rangle$ describes known projections of the two components. In the coupled basis $|JM\rangle$, the magnitude of the *total* angular momentum is known. The relation is a change of basis:

$$\boxed{|JM\rangle = \sum_{m_1 m_2} \langle j_1 m_1, j_2 m_2 | JM\rangle \, |j_1 m_1\rangle \, |j_2 m_2\rangle \, .}$$

The number in angle brackets is a Clebsch–Gordan (CG) coefficient. It is a probability *amplitude* for a particular uncoupled pair, not a physical coupling energy or a probability by itself.

Worked example: Two spin-$\frac{1}{2}$ particles from scratch

Each particle has $|\uparrow\rangle$ or $|\downarrow\rangle$. Addition allows $J = 1$ (triplet) and $J = 0$ (singlet). The coupled states are

$$\begin{aligned} |1,1\rangle &= |\uparrow\uparrow\rangle , \\ |1,0\rangle &= \frac{|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle}{\sqrt{2}}, \\ |1,-1\rangle &= |\downarrow\downarrow\rangle , \\ |0,0\rangle &= \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \end{aligned}$$

Thus $\langle \frac{1}{2}, \frac{1}{2}; \frac{1}{2}, -\frac{1}{2} | 1, 0\rangle = 1/\sqrt{2}$. The same product basis component has overlap $1/\sqrt{2}$ with the singlet. Triplet and singlet states are orthogonal, and the four coupled states span the same four-dimensional Hilbert space as the four product states.

15.3 Why introduce a 3-j symbol?

A Wigner 3-j symbol is a conventional normalised version of a CG coefficient:

$$\boxed{\begin{pmatrix} j_1 & j_2 & J \\ m_1 & m_2 & -M \end{pmatrix} = \frac{(-1)^{j_1 - j_2 + M}}{\sqrt{2J+1}} \langle j_1 m_1, j_2 m_2 | JM\rangle .}$$

It is useful because its symmetry properties make selection rules and angular integrations compact. Before calculating a value, test:

$$|j_1 - j_2| \le j_3 \le j_1 + j_2, \qquad m_1 + m_2 + m_3 = 0, \qquad |m_i| \le j_i .$$

If a condition fails, the symbol is zero. This often answers a whole transition question before numerical evaluation.

Worked example: A concrete $3j$ calculation

Adding zero angular momentum to $j = 1$ does nothing:

$$\langle 0, 0; 1, 0 | 1, 0 \rangle = 1.$$

Use the relation above with $j_1 = 0, j_2 = 1, J = 1, M = 0$:

$$\begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \frac{(-1)^{0-1+0}}{\sqrt{3}}(1) = \boxed{-\frac{1}{\sqrt{3}}}.$$

There is no need to memorise the full factorial formula for this special case. For less obvious values, consult the CG/3-j tables provided in the course or use a symbolic-algebra program.

15.4 Wigner–Eckart: separate geometry from physics

A spherical tensor operator $T_q^{(k)}$ has rank k and component $q = -k, \ldots, k$. Electric dipole is rank 1; electric quadrupole is rank 2. For a chosen reduced-element convention,

$$\boxed{\left\langle j'm' \middle| T_q^{(k)} \middle| jm \right\rangle = (-1)^{j'-m'} \begin{pmatrix} j' & k & j \\ -m' & q & m \end{pmatrix} \langle j' \| T^{(k)} \| j \rangle.}$$

The 3-j symbol gives the allowed *geometry*, including the projection condition $m' = m + q$. The reduced matrix element contains the specific radial/atomic physics but is independent of m, m', q. Two transitions within the same multiplet can therefore have different strengths entirely because their 3-j coefficients differ.

15.5 A transition example with $J = 0 \to J' = 1$

An E1 operator has $k = 1$. Begin in $J = 0, m = 0$. The triangle rule permits $J' = 1$. If the light has tensor component $q = +1$, then $m' = 1$ because $m' = m + q$. For $q = 0$, the only final projection is $m' = 0$. In both cases the absolute angular coefficient is $1/\sqrt{3}$ in the displayed Wigner–Eckart convention. Whether the actual transition occurs still depends on the *reduced* dipole element and parity.

15.6 Why parity is an independent test

Under spatial inversion, $\mathbf{r} \mapsto -\mathbf{r}$. A dipole operator $\mathbf{d} = -e\mathbf{r}$ changes sign. If the initial and final states have identical parity, the integrand of $\langle f|\mathbf{d}|i\rangle$ is odd and the integral vanishes. Therefore a nonzero 3-j symbol does *not* by itself guarantee E1 emission.

Concept check

Use the shortest sequence: (1) determine total J and projection possibilities; (2) test 3-j triangle and magnetic-number rules; (3) check parity; (4) only then evaluate a 3-j coefficient and reduced atomic matrix element.

16 Fine structure, hyperfine structure and fields

Course sources: Weeks 8–9; Assignment 3 Q4; 2025 Q1

16.1 Why one electronic configuration has many levels

The central-field approximation produces orbital energies but does not yet include all small interactions. Electron spin–orbit coupling links the orbital and spin magnetic moments; the atomic nucleus has its own spin and magnetic moment; external electric and magnetic fields shift the levels. These effects create fine, hyperfine, Zeeman and Stark structures with different typical energy scales.

16.2 Fine structure from $\mathbf{L}\cdot\mathbf{S}$

Suppose $H_{\mathrm{SO}} = A_{\mathrm{SO}}\,\mathbf{L}\cdot\mathbf{S}/\hbar^2$ within an LS term. Since

$$\mathbf{J}^2 = (\mathbf{L}+\mathbf{S})^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L}\cdot\mathbf{S},$$

the energy correction is

$$\boxed{E_J^{\mathrm{SO}} = \frac{A_{\mathrm{SO}}}{2}[J(J+1) - L(L+1) - S(S+1)].}$$

This is more than a formula to remember: it is simply the scalar product identity for two angular momenta.

Worked example: Split a ^3D term

The term has $S=1, L=2$, so $J=1,2,3$. With $L(L+1)=6$ and $S(S+1)=2$,

$$E_1 = -3A_{\mathrm{SO}}, \quad E_2 = -A_{\mathrm{SO}}, \quad E_3 = 2A_{\mathrm{SO}}.$$

The degeneracies $3+5+7=15$ equal $(2L+1)(2S+1) = 5\times 3$. The level intervals $2A_{\mathrm{SO}}$ and $3A_{\mathrm{SO}}$ are not equally spaced.

16.3 Hyperfine structure couples electrons to the nucleus

The nuclear spin $\mathbf{I}$ combines with electronic $\mathbf{J}$ to form $\mathbf{F} = \mathbf{I}+\mathbf{J}$. For magnetic-dipole hyperfine coupling,

$$H_{\mathrm{hfs}} = \frac{A}{\hbar^2}\mathbf{I}\cdot\mathbf{J},$$

giving

$$\boxed{E_F = \frac{A}{2}[F(F+1) - I(I+1) - J(J+1)]}, \quad F = |I-J|, \ldots, I+J.$$

The numerical sign of A matters: the labels alone do not establish which F has higher energy. Electric quadrupole and higher-rank nuclear hyperfine interactions are possible only if the nuclear and electronic tensor ranks allow them.

Worked example: The 2025 exam hyperfine example

Let $I = 5/2, J = 1/2$. Then $F = 2, 3$. Since $I(I+1) = 35/4$, $J(J+1) = 3/4$,

$$E_2 = \frac{A}{2}(6 - 38/4) = \boxed{-7A/4}, \quad E_3 = \frac{A}{2}(12 - 38/4) = \boxed{5A/4}.$$

Their separation is $3|A|$. The magnetic substate count $(2F_1+1) + (2F_2+1) = 5+7 = 12$ agrees with $(2I+1)(2J+1) = 6\times 2$.

16.4 The weak magnetic-field regime

A magnetic field interacts with atomic magnetic moments. When hyperfine coupling dominates the Zeeman energy, F, m_F remain useful approximate quantum numbers:

$$\boxed{\Delta E_{Fm_F} \simeq \mu_B g_F m_F B.}$$

Neglecting the much smaller nuclear magnetic contribution and for $F > 0$,

$$g_F \simeq g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.$$

The electronic Landé factor for $g_L = 1, g_S \simeq 2$ is

$$\boxed{g_J \simeq 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}}.$$

These factors arise from *projecting* electronic orbital/spin moments first onto J, then onto F.

16.5 The strong-field (Paschen–Back) regime

When Zeeman energy exceeds hyperfine coupling, nuclear and electronic angular momenta approximately decouple. The natural basis changes from $|Fm_F\rangle$ to $|Jm_J, Im_I\rangle$. To first order,

$$E \simeq g_J \mu_B m_J B + A m_I m_J$$

if the nuclear Zeeman contribution is neglected. The total state count stays constant, but the interpretation of the levels changes. A fixed-F weak-field formula must *not* be extrapolated to arbitrarily large B.

16.6 Electric fields: the Stark effect

A static electric field $\mathcal{E}\hat{z}$ couples through $-\mathbf{d}\cdot\boldsymbol{\mathcal{E}}$. For a nondegenerate parity eigenstate with no permanent electric dipole, the first-order shift vanishes. Second-order perturbation theory gives

$$\boxed{\Delta E^{(2)} = -\frac{1}{2}\alpha(0)\mathcal{E}^2, \quad \alpha(0) = 2\sum_{n\neq i} \frac{|\langle n|d_z|i\rangle|^2}{E_n - E_i}}.$$

The polarisability measures how readily the atom's electron cloud responds to an applied field. For a hydrogenic $1s$ state a closure approximation using $E_n - E_1 \approx E_2 - E_1$ gives $\alpha_{\text{approx}} = 16/(3Z^4)$ in atomic polarizability-volume units, compared with the exact hydrogen value $9/2$ at $Z = 1$.

Concept check

Identify the relevant coupling hierarchy. Fine structure: $L+S \to J$. Hyperfine: $I+J \to F$. Weak magnetic field: F, m_F. Strong magnetic field: m_I, m_J. Electric field: check parity before applying Stark perturbation theory.

17 Atomic spectroscopy, multipoles and radiative lifetimes

Course sources: Week 9; Week 10; Assignment 4 Q3

17.1 Absorption and emission connect energy levels

An atom absorbs light when an electromagnetic mode drives a transition between its quantum energy eigenstates. Energy conservation fixes

$$\hbar\omega_{if} = E_i - E_f, \quad \nu = \frac{E_i - E_f}{h}, \quad \lambda = \frac{hc}{E_i - E_f}.$$

A positive energy difference alone does not guarantee a transition: the light–matter interaction operator must have a nonzero matrix element between the two states.

17.2 Electric dipole coupling

The leading long-wavelength interaction is $H_{\text{int}} = -\mathbf{d}\cdot\mathbf{E}$, where the electron dipole is $\mathbf{d} = -e\mathbf{r}$. The transition amplitude contains

$$\mathcal{M}_{fi} \propto \langle f|\,\mathbf{d}\cdot\boldsymbol{\epsilon}\,|i\rangle\,.$$

This separates into radial and angular factors when orbitals are expressed as $R_{nl}(r)Y_{lm}(\theta,\phi)$. For an electric dipole the radial integral is

$$\boxed{\int_0^\infty R_f^*(r)R_i(r)\,r^3 dr}$$

or, using $u = rR$, $\int u_f^*(r)\,r\,u_i(r)dr$. The power r^3 combines the r^2dr volume measure with the additional r from the dipole.

17.3 Multipole rank and parity rules

An electric multipole Ek is a rank-k spherical tensor of parity $(-1)^k$; a magnetic multipole Mk has parity $(-1)^{k+1}$. For total angular momenta J, J', the geometric rule is

$$\boxed{|J - J'| \le k \le J + J', \qquad m' = m + q, \quad |q| \le k.}$$

For the most common transitions:

	Parity	Common J rules
$E1$ $(k=1)$	changes	$\Delta J = 0, \pm 1,\ 0 \not\leftrightarrow 0$
$M1$ $(k=1)$	same	$\Delta J = 0, \pm 1,\ 0 \not\leftrightarrow 0$
$E2$ $(k=2)$	same	$\lvert\Delta J\rvert \le 2,\ J + J' \ge 2$

The usual E1 orbital rule for one electron is $\Delta l = \pm 1$. For E2 the single-electron orbital rules are $\Delta l = 0, \pm 2$, subject to triangle conditions. Thus $s \leftrightarrow d$ can be E2, but $s \leftrightarrow p$ is not E2 because parity changes. Additional approximate LS coupling rules, such as $\Delta S = 0$ for electric-dipole transitions, may be softened by spin–orbit mixing. A weak intercombination line can therefore exist even if it is forbidden in ideal LS coupling.

Worked example: Which sodium transitions can radiate by E1?

The valence-orbital transitions $3p \to 3s$, $4s \to 3p$ and $3d \to 3p$ all have $\Delta l = \pm 1$, so their orbital E1 matrix elements may be nonzero. But $3d \to 3s$ has $\Delta l = -2$ and is E1 forbidden. Always check *downward energy ordering* for spontaneous emission: an angularly allowed upward transition is absorption, not spontaneous decay from a lower level.

17.4 Why some states decay quickly

An excited atom couples to vacuum electromagnetic modes and can spontaneously emit a photon. For an E1 transition between appropriately specified states,

$$\boxed{A_{if} = \frac{\omega_{if}^3}{3\pi\epsilon_0\hbar c^3} |\langle f| \mathbf{d} |i\rangle|^2,}$$

where the displayed vector dipole strength and sublevel/polarisation summations must be handled consistently. The strong ω^3 dependence means larger energy gaps generally radiate faster for comparable matrix elements. A forbidden E1 line may instead decay by much weaker M1/E2 channels, making it useful for long-lived clock or metastable states.

17.5 Lifetime and branching ratios

If an excited state can decay into several lower states,

$$\boxed{\Gamma_i = \sum_{f<i} A_{if}, \quad \tau_i = \Gamma_i^{-1}, \quad b_{if} = \frac{A_{if}}{\Gamma_i}.}$$

Do *not* use the inverse of just one A if multiple important decay channels exist. In quantum optics Γ_i is a population decay rate with units s^{-1}; a natural linewidth in angular-frequency units is of order Γ_i.

Worked example: A transition-rate calculation with a stated dipole

For a *hypothetical* 589 nm line with $|\mathbf{d}| = 2ea_0$ and no other downward channels,

$$\omega = \frac{2\pi c}{589\,\mathrm{nm}}, \quad A = \frac{\omega^3 (2ea_0)^2}{3\pi\epsilon_0\hbar c^3}.$$

This gives a rate of order $4 \times 10^7\ \mathrm{s}^{-1}$ and a lifetime of order 25 ns. It is an illustration of using the formula, *not* a claim about a measured Na lifetime. For actual sodium, compute the wavefunction-derived matrix elements and sum all allowed channels.

17.6 Einstein coefficients and induced processes

An incident radiation field can stimulate both absorption and emission. Einstein's A coefficient describes spontaneous decay; the B coefficients describe rates proportional to the radiation energy density. Their detailed-balance relations explain thermal equilibrium radiation and underlie laser gain calculations. A large population inversion makes stimulated emission dominate absorption at the selected lasing transition.

Concept check

A complete transition question requires *energy*, *tensor rank*, *angular-momentum triangle*, *parity*, and a *reduced matrix element*. A lifetime then needs all allowed downward branches.

18 Driving, cooling and trapping atoms with lasers

Course sources: Week 10; 2025 Q2; Zeeman slower tutorial

18.1 A laser couples two atomic levels

Near an isolated optical transition, model the atom with $|g\rangle, |e\rangle$. A classical electric field $\mathbf{E}(t) = \mathbf{E}_0 \cos(\omega_L t)$ couples through $-\mathbf{d} \cdot \mathbf{E}(t)$. The *Rabi frequency* measures interaction strength:

$$\boxed{\Omega = \frac{|\langle e| \mathbf{d} \cdot \mathbf{E}_0 |g\rangle|}{\hbar}}$$

with factors of 2 depending on whether E_0 denotes the full real-field amplitude or its positive-frequency part. State the convention in a pulse-timing problem. In a rotating frame and under the rotating-wave approximation, a convenient effective Hamiltonian is

$$H_{\text{rot}} = \frac{\hbar}{2} \begin{pmatrix} -\Delta & \Omega \\ \Omega & \Delta \end{pmatrix}, \quad \Delta = \omega_L - \omega_{eg}.$$

The generalised Rabi frequency is $\Omega_{\text{eff}} = \sqrt{\Omega^2 + \Delta^2}$.

18.2 Pulse area, pi and pi-over-two pulses

For a resonant pulse with constant coupling phase and time-dependent amplitude $\Omega(t)$, the pulse area is $\Theta = \int \Omega(t) dt$. An atom starting in $|g\rangle$ has

$$\boxed{P_e = \sin^2(\Theta/2).}$$

A π-pulse transfers all population to the excited state in the ideal lossless model; a $\pi/2$-pulse creates an equal-amplitude superposition. Detuning prevents perfect inversion for a constant-frequency rectangular pulse, because the largest excited probability is $\Omega^2/(\Omega^2 + \Delta^2)$.

18.3 Why light can exert a force

A photon carries momentum $\hbar\mathbf{k}$. Absorption transfers this directed momentum to an atom. Spontaneous photons are emitted in different directions, so their average momentum along a fixed axis may vanish, though their variance gives recoil heating. For a driven two-level atom with excited-state decay rate Γ, a common steady-state scattering rate is

$$\boxed{R_{\text{sc}} = \frac{\Gamma}{2} \frac{s}{1 + s + (2\Delta/\Gamma)^2}}, \qquad s = I/I_{\text{sat}}.$$

At high saturation on resonance, $R_{\text{sc}} \to \Gamma/2$, giving

$$\boxed{F_{\max} = \hbar k \Gamma/2, \qquad a_{\max} = F_{\max}/M.}$$

There is a trade-off: fast-scattering broad transitions give strong slowing forces but generally higher Doppler cooling limits than narrow lines.

18.4 Optical molasses and the Doppler shift

An atom moving toward one of two counter-propagating red-detuned beams sees that beam Doppler-shifted closer to resonance. It absorbs more photons from the direction opposing its motion, creating a damping force near rest, $F \simeq -\beta v$. In a simple two-level Doppler cooling model, the minimum temperature is

$$\boxed{T_D = \frac{\hbar\Gamma}{2k_B} = \frac{\hbar}{2k_B\tau}},$$

where $\tau = 1/\Gamma$. The temperature comes from balancing velocity damping against random photon-recoil diffusion; the kinetic energy never falls to exactly zero.

18.5 A Zeeman slower follows changing velocity

A fast atomic beam has a large Doppler shift, which changes as atoms slow. A position-dependent magnetic field Zeeman-shifts the atomic transition to maintain resonance with the counter-propagating laser. If the designed deceleration is constant,

$$v(z) = \sqrt{v_0^2 - 2az}, \quad L = \frac{v_0^2}{2a},$$

and, for a suitable sign convention, the resonance field has the shape

$$\boxed{B(z) = B_{\text{offset}} + B_0\sqrt{1 - z/L}.}$$

This is the origin of the characteristic square-root slower profile, not merely a curve fitted to experimental data. An arbitrary field that reaches zero need not slow atoms successfully: the corresponding resonant velocity must be dynamically followable. Specifically,

$$\boxed{a_{\text{req}}(z) = -v_{\text{res}}(z)\frac{dv_{\text{res}}}{dz}}$$

must stay within the available scattering acceleration and capture linewidth.

Worked example: Estimate ytterbium stopping distance

For the 2025 exam data, $M = 1.73 \times 10^{-25}\,\text{kg}$, $T = 300\,\text{K}$, $\lambda = 399\,\text{nm}$, $\tau = 5.5\,\text{ns}$, choose the 3D rms speed $v_0 = \sqrt{3k_BT/M} \approx 268\,\text{m/s}$. At half the maximum scattering force,

$$a = \frac{h}{4M\lambda\tau} \approx 4.36 \times 10^5\,\text{m/s}^2.$$

Hence

$$L = \frac{v_0^2}{2a} \approx 8.23\,\text{cm}, \qquad N_\gamma = \frac{Mv_0\lambda}{h} \approx 2.79 \times 10^4.$$

The Doppler limit from this linewidth is $T_D \approx 0.694\,\text{mK}$. The thermal speed convention affects the stopping distance; state it explicitly.

18.6 Magneto-optical traps and selection rules

A magneto-optical trap (MOT) uses opposing red-detuned laser beams with suitable circular polarisations and a magnetic-field gradient. Position-dependent Zeeman shifts make one opposing beam more resonant, producing a restoring force, while the Doppler effect supplies damping. Polarisation selection rules connect light to atomic magnetic substates: π-polarised light has $\Delta m = 0$; $\sigma^{\pm}$ components have $\Delta m = \pm 1$, for a consistent quantisation axis and photon helicity. Optical pumping can redistribute population among those substates.

Concept check

Rabi oscillations describe coherent drive. Spontaneous scattering describes dissipative momentum kicks. Doppler molasses damps velocity; a MOT also restores position. Zeeman slowing maintains resonance while the beam loses speed.

19 Atoms as tests of parity and time-reversal symmetry

Course sources: Week 12; atomic parity violation and EDM lectures

19.1 Why symmetry is a practical experimental tool

An atomic experiment can be sensitive to tiny interactions because ordinary electromagnetic theory predicts that certain transitions or energy shifts should be *zero* by symmetry. If a supposedly forbidden signal appears in a controlled measurement, we investigate whether an interaction absent from the simpler Hamiltonian has broken that symmetry. This motivates atomic tests of the weak interaction, parity violation and intrinsic electric dipole moments.

19.2 What parity means

A parity transformation inverts the spatial coordinates: $\mathbf{r} \mapsto -\mathbf{r}$. For a central potential, the orbital wavefunctions have definite parity:

$$P\,|nlm\rangle = (-1)^l\,|nlm\rangle\,.$$

Thus s, d, g orbitals are even; p, f orbitals are odd. The electric dipole operator $\mathbf{d} = -e\mathbf{r}$ is itself odd under parity. If initial and final states have the *same* parity,

$$\langle f|\,\mathbf{d}\,|i\rangle = -\,\langle f|\,\mathbf{d}\,|i\rangle = 0$$

in a parity-conserving model. This explains the E1 parity selection rule without memorising a table.

19.3 How the weak interaction changes the conclusion

The electromagnetic Hamiltonian of an atom approximately respects parity, but the neutral-current weak interaction contains parity-violating electron–nucleus terms. Schematically,

$$H = H_{\mathrm{EM}} + H_W, \quad PH_WP^{-1} \neq H_W.$$

First-order perturbation theory shows that an originally definite-parity state $|i\rangle$ acquires admixtures of opposite-parity states:

$$\boxed{\left|\tilde{i}\right\rangle \approx |i\rangle + \sum_{n\neq i} |n\rangle\,\frac{\langle n|\,H_W\,|i\rangle}{E_i - E_n}.}$$

These admixtures can generate an otherwise forbidden E1 amplitude between same-parity levels.

Worked example: Derive a weakly allowed E1 line

Let $|i\rangle\,, |f\rangle$ have the same unperturbed parity and $|n\rangle$ the opposite parity. If only the initial-state admixture matters,

$$\left|\tilde{i}\right\rangle = |i\rangle + \epsilon\,|n\rangle\,, \quad \epsilon = \frac{\langle n|\,H_W\,|i\rangle}{E_i - E_n}.$$

Then

$$\langle f|\,d\left|\tilde{i}\right\rangle = \underbrace{\langle f|\,d\,|i\rangle}_{0} + \epsilon\,\langle f|\,d\,|n\rangle = \boxed{\epsilon\,\langle f|\,d\,|n\rangle}.$$

A small mixing coefficient 10^{-8} multiplied by an allowed atomic dipole of order ea_0 gives a $10^{-8}ea_0$ amplitude. In a full calculation, both initial and final states can be mixed, and one sums over many intermediate states.

19.4 Nuclear weak charge

One widely discussed contribution to atomic parity violation is controlled by a nuclear weak charge, with tree-level approximate form

$$\boxed{Q_W \simeq -N + Z(1 - 4\sin^2\theta_W),}$$

where N, Z are neutron and proton numbers. Electroweak radiative corrections and nuclear-structure effects matter in precision work. The measured parity-violating amplitude also depends strongly on the atomic electronic wavefunctions, so comparing experiment with the Standard Model requires accurate atomic many-body theory, not just the weak charge.

19.5 What is an intrinsic electric dipole moment?

An electric dipole $\mathbf{d}$ changes sign under parity, while an axial angular momentum $\mathbf{J}$ does not. A permanent $\mathbf{d}$ aligned with a particle's spin therefore violates parity. Under time reversal, spin changes direction but the electric dipole does not, so such an intrinsic EDM also violates time reversal. The interaction energy is

$$\boxed{H_{\mathrm{EDM}} = -\mathbf{d}\cdot\boldsymbol{\mathcal{E}}.}$$

A spin-precession experiment compares frequencies for opposite electric-field directions. A genuine EDM contribution reverses sign with the electric field, while carefully chosen combinations of measurements reject common magnetic and instrumental backgrounds.

19.6 Schiff screening and experimental enhancement

In a simple neutral atom of pointlike, nonrelativistic Coulomb-interacting constituents, internal fields screen a uniformly applied static electric field at the nucleus. This is the basis of Schiff's theorem. Finite nuclear size, relativistic electron motion and molecular polarisation circumvent its ideal assumptions and can make EDM measurements extremely sensitive. Heavy atoms and polar molecules are particularly useful because their electronic and nuclear structure can amplify some symmetry-violating effects.

19.7 Why this belongs in atomic physics

The experimental quantity is a tiny frequency shift or transition amplitude. To predict it, one still needs the atomic orbitals, good angular-momentum coupling, selection rules, perturbation theory and matrix elements developed earlier in this mini-text. A precision experiment turns those familiar mathematical tools into a test of fundamental interactions.

Concept check

Parity selection rules forbid amplitudes *only if* the Hamiltonian respects parity and the basis states have definite parity. A weak mixing interaction changes the state and may open a previously vanishing transition.

20 Atomic clocks and precision spectroscopy

Course sources: Week 13; 2025 Q1; atomic applications

20.1 What makes an atom a clock?

A clock is an oscillator whose phase can be compared against a reference. An atomic transition provides a frequency associated with an exceptionally stable energy difference:

$$\boxed{\nu_0 = \frac{E_{\text{upper}} - E_{\text{lower}}}{h}.}$$

A laser or microwave oscillator is steered to keep it on resonance with this frequency. The atom does not generate the counted electrical clock pulses by itself: an interrogation sequence and feedback system are needed.

20.2 The caesium definition of the SI second

The unperturbed ground-state hyperfine transition of ^{133}Cs defines the second, with frequency

$$\boxed{\nu_{\text{Cs}} = 9\,192\,631\,770 \text{ Hz}.}$$

Caesium's valence ground term is $6s\,^2S_{1/2}$, so $J = 1/2$, while the nuclear spin is $I = 7/2$. Hyperfine coupling gives $F = 3, 4$. The clock transition is the

$$\boxed{|F = 3, m_F = 0\rangle \longleftrightarrow |F = 4, m_F = 0\rangle}$$

transition. The choice $m_F = 0$ makes the first-order Zeeman shift vanish near zero field, reducing a major source of frequency error.

Worked example: Compute the caesium hyperfine interval

With $I = 7/2, J = 1/2$, $I(I+1) = 63/4$ and $J(J+1) = 3/4$. The magnetic-dipole hyperfine energy is

$$E_F = \frac{A}{2}[F(F+1) - I(I+1) - J(J+1)].$$

Substitute $F = 3, 4$:

$$E_3 = \frac{A}{2}(12 - 66/4) = -\frac{9A}{4}, \qquad E_4 = \frac{A}{2}(20 - 66/4) = \frac{7A}{4}.$$

Therefore $E_4 - E_3 = 4A$, so

$$\boxed{A/h = \nu_{\text{Cs}}/4 = 2.2981579425 \text{ GHz}.}$$

The degeneracy check is $7 + 9 = 16 = (2I+1)(2J+1)$.

20.3 Ramsey spectroscopy: letting phase accumulate

In a Ramsey experiment a first $\pi/2$ pulse prepares an equal superposition of clock states. The atomic components then acquire a relative phase during free evolution for time T. A second $\pi/2$ pulse converts that phase into population. In the idealised two-level model, the excited population oscillates with detuning:

$$P_e \sim \tfrac{1}{2}[1 + \cos(\Delta T + \phi_0)].$$

The fringe width scales approximately as $1/T$. Long coherent interrogation therefore permits a narrow frequency discriminator, provided the atom and the oscillator stay coherent.

20.4 Why optical clocks can do better

The quality factor of a spectral reference is

$$\boxed{Q = \frac{\nu_0}{\Delta\nu}}.$$

A high carrier frequency and narrow linewidth can yield a very high Q. For example, strontium optical clocks use a transition near 429 THz, vastly higher than the 9.19 GHz caesium hyperfine carrier. Optical clocks usually use specially chosen metastable, narrow transitions rather than bright, rapidly decaying laser-cooling lines.

20.5 Trapping without shifting the clock

An optical lattice localises atoms, reducing motional Doppler effects. However, trapping light also shifts atomic energy levels through the AC Stark effect. At a *magic wavelength*, the differential light shift of the two clock states is approximately zero, permitting confinement without substantially shifting the measured transition frequency.

20.6 Stability and accuracy are different

Stability describes the scatter of clock frequency measurements over an averaging time; accuracy describes how closely the corrected frequency matches the unperturbed atomic reference. Quantum projection noise, technical oscillator noise and finite atom number affect stability. Zeeman/Stark shifts, black-body radiation, collisions and gravitational redshift are among the corrections important for accuracy. A clock can be very stable yet systematically wrong.

20.7 Why clocks are physics experiments

Comparing clocks based on different species or transitions provides tests of relativity and searches for changes in fundamental constants. Relativistic gravitational redshift means clocks at different gravitational potentials tick at measurably different rates. Precise atomic spectroscopy therefore connects quantum structure to timekeeping, geodesy and fundamental physics.

Concept check

A clock should have a long-lived, narrow transition and be insensitive to external perturbations. The atomic transition, interrogation sequence, local oscillator, feedback and systematic corrections all determine clock performance.

21 A practical mathematical toolkit

Course sources: Common methods across all four course parts

21.1 Expectation values in an eigenbasis

Suppose $|\psi\rangle = \sum_n c_n |n\rangle$, where $|n\rangle$ are eigenstates of an observable A with eigenvalues a_n. Then

$$\langle A\rangle = \sum_n |c_n|^2 a_n, \quad (\Delta A)^2 = \sum_n |c_n|^2 a_n^2 - \left(\sum_n |c_n|^2 a_n\right)^2.$$

The *probabilities* are $|c_n|^2$, not c_n. For operators not diagonal in this basis, retain cross terms $c_m^* c_n \langle m| A |n\rangle$.

21.2 Commutators and transformations

For Heisenberg evolution $\dot{O} = (i/\hbar)[H, O]$. If H is a two-level Hamiltonian $H = (\hbar\Omega/2)\sigma_x$,

$$U(t) = e^{-i\Omega t\sigma_x/2} = I\cos(\Omega t/2) - i\sigma_x \sin(\Omega t/2),$$

because $\sigma_x^2 = I$. For field displacement $D = e^{\alpha a^\dagger - \alpha^* a}$, the identity $D^\dagger a D = a + \alpha$ follows by truncating the nested commutator expansion after one step.

21.3 Useful sums and integrals

The geometric series for $|q| < 1$ is $\sum_{n=0}^{\infty} q^n = (1-q)^{-1}$. Differentiate with respect to q to obtain

$$\sum_{n=0}^{\infty} nq^n = \frac{q}{(1-q)^2}, \quad \sum_{n=0}^{\infty} n^2 q^n = \frac{q(1+q)}{(1-q)^3}.$$

For $\beta > 0$ and integer $m \geq 0$,

$$\boxed{\int_0^\infty r^m e^{-\beta r} dr = \frac{m!}{\beta^{m+1}}.}$$

These formulas appear in thermal photon statistics and hydrogen radial expectation calculations.

21.4 Perturbation theory

For $H = H_0 + \lambda V$ and a nondegenerate unperturbed state $|i\rangle$,

$$E_i^{(1)} = \langle i| V |i\rangle, \qquad E_i^{(2)} = \sum_{n\neq i} \frac{|\langle n| V |i\rangle|^2}{E_i^{(0)} - E_n^{(0)}}.$$

A first-order correction to the *state* is

$$\left|i^{(1)}\right\rangle = \sum_{n\neq i} |n\rangle \frac{\langle n| V |i\rangle}{E_i^{(0)} - E_n^{(0)}}.$$

Use the energy formula for a finite-nucleus or Stark *shift*; use the state-mixing formula to explain how weak parity violation creates a forbidden transition. For degenerate levels, first diagonalise the perturbation in the degenerate subspace.

21.5 Fourier transforms of differential equations

With the convention $\tilde{f}(\omega) = \int f(t) e^{-i\omega t} dt$, $\widetilde{\dot{f}} = i\omega \tilde{f}$. Hence

$$\dot{a} = -\lambda a + f(t) \quad \Longrightarrow \quad \boxed{\tilde{a}(\omega) = \frac{\tilde{f}(\omega)}{\lambda + i\omega}}.$$

A cavity susceptibility is an instance of this elementary linear-response formula. The real damping rate appears in the denominator together with frequency detuning.

21.6 Normalisation of conditional quantum states

A detected branch is generally $\tilde{\rho} = K\rho K^\dagger$. Keep its trace as the success probability $p = \operatorname{Tr}\tilde{\rho}$ and only then write $\rho_{\text{cond}} = \tilde{\rho}/p$. It is mathematically invalid to normalise individual terms before adding coherent amplitudes or to sum conditional probabilities without their branch weights.

Worked example: A quick selection-rule test

Suppose a putative E2 transition goes from $J = 0$ to $J' = 1$. E2 has tensor rank $k = 2$. The triangle rule requires

$$|0 - 1| \leq 2 \leq 0 + 1,$$

which fails at the right-hand inequality. Therefore the rank-2 matrix element vanishes before any radial integral is evaluated. A parity check is still needed for other pairs that pass the triangle rule.

Concept check

Often the difficult part is recognising the *right representation*. Fock expansions for photon counts; quadratures for homodyne/squeezing; Kraus operators for channels; small angular-momentum blocks for atomic splittings; reduced matrix elements for spectral transitions.

22 Question-type atlas, glossary and further practice

Course sources: All tutorials, assignments and 2023–2025 examinations

22.1 Recognise a question before reaching for a formula

The same few strategies recur throughout this course. Use the table below to find the relevant teaching chapter and then consult the longer worked-solutions booklets.

If the question asks about...	Start by doing this
Coherent / thermal number counts	Expand in $\|n\rangle$, form P_n, evaluate mean and factorial moments (Chapters 1–2)
HOM or photonic circuits	Transform *creation* operators and normalise Fock states (Chapter 3)
Quadratures / squeezing	State $[x, p] = 2i$; compute means and covariance, then apply the noise model (4–5)
Bell protocols and QKD	Write the actual input state, gates, measurement branches and sifted conditions (6–7)
Optical loss / postselection	Include an environmental mode, trace or project and normalise correctly (8–10)
Langevin cavity	Commute Hamiltonian, Fourier transform and apply the input–output boundary relation (9)
Atom–cavity coupling	Choose conserved-excitation basis and diagonalise a small block (11)
Quantum metrology	Identify parameter generator and actual measurement probabilities (12)
Hydrogen / nuclear size	Separate $R(r)Y_{lm}$; verify the radial measure and use perturbation theory (13)
Configurations / LS terms	Enforce antisymmetry; couple L, S, J; check the total state count (14)
3-j / transition amplitude	Check triangle and m sums, then Wigner–Eckart and parity (15,17)
Fine / hyperfine / Zeeman	Choose L, S, J, I, F, identify weak- or strong-field basis, compute splitting (16)
Lifetime / multipoles	Select allowed tensor rank and parity, calculate dipole, sum all downward A's (17)
Rabi pulse / laser cooling	Distinguish coherent pulse area from dissipative scattering force (18)
Atomic PV / EDM	Find the symmetry-forbidden leading term, then perturbatively mix states (19)
Cesium / optical clock	Calculate hyperfine reference, identify field-insensitive sublevels and systematic shifts (20)

22.2 Frequently confused concepts

Probability amplitude vs probability: a quantum state uses complex amplitudes; only their modulus squared gives a probability.

Coherent state vs pure state: a coherent state is one particular pure oscillator state; many other pure states are nonclassical.

Mode overlap vs optical power: photon interference visibility depends on mode indistinguishability,

not just intensity.

Relative phase vs global phase: a global factor $e^{i\phi}\,|\psi\rangle$ does not change measurement results, but a relative phase between components does.

Kraus channel vs conditional outcome: summed branches are trace preserving; one selected detector outcome must be renormalised and assigned a success probability.

Angular momentum J vs term L, S: total electronic J is the vector sum of already coupled electronic orbital and spin moments. Hyperfine F then adds nuclear I.

Allowed angular transition vs nonzero line strength: a 3-j symbol can pass selection rules while the reduced atomic matrix element vanishes.

Natural linewidth vs laser detuning: linewidth arises from decay/decoherence; detuning is the frequency mismatch between laser and atomic resonance.

Clock accuracy vs stability: accuracy concerns systematic offset; stability concerns fluctuations over averaging time.

22.3 Physical constants and units that recur

$\hbar = h/(2\pi)$	reduced Planck constant
$a_0 = 5.29177 \times 10^{-11}\,\mathrm{m}$	Bohr radius
$1E_h = 27.2114\,\mathrm{eV}$	Hartree energy
$1E_h/(hc) = 219474.63\,\mathrm{cm}^{-1}$	atomic spectroscopy
$t_{\mathrm{au}} = 2.418884 \times 10^{-17}\,\mathrm{s}$	atomic unit of time
$\mu_B = e\hbar/(2m_e)$	Bohr magneton
$\alpha_{\mathrm{fs}} \simeq 1/137.036$	fine-structure constant

When a problem switches between wavelength, wavenumber, frequency and angular frequency, write each definition explicitly:

$$\tilde{\nu} = 1/\lambda, \quad \nu = c/\lambda, \quad \omega = 2\pi\nu, \quad E = h\nu = \hbar\omega.$$

The factors of 2π are particularly important in linewidth and Rabi calculations.

22.4 Where to practise each technique

The repository companion booklets are designed to work together:

1. The *15-page Course Overview* is the fast exam-day map of key physical results.
2. The *Worked Examples* booklet provides twenty carefully selected full calculations.
3. The *Assignment Solutions* booklet develops derivations and numerical methods associated with four assignment sheets.
4. The *2023–2025 Past Exam Solutions* booklet covers twelve full exam questions.

In revision, read a mini-text chapter and try the associated problem *before* looking at its worked solution. List any remaining conceptual difficulty in the repository's mistakes-and-gaps notes.

Concept check

An exam solution is strongest when it connects a choice of model, a governing operator, a calculation, a limiting-case check and a physical interpretation.